

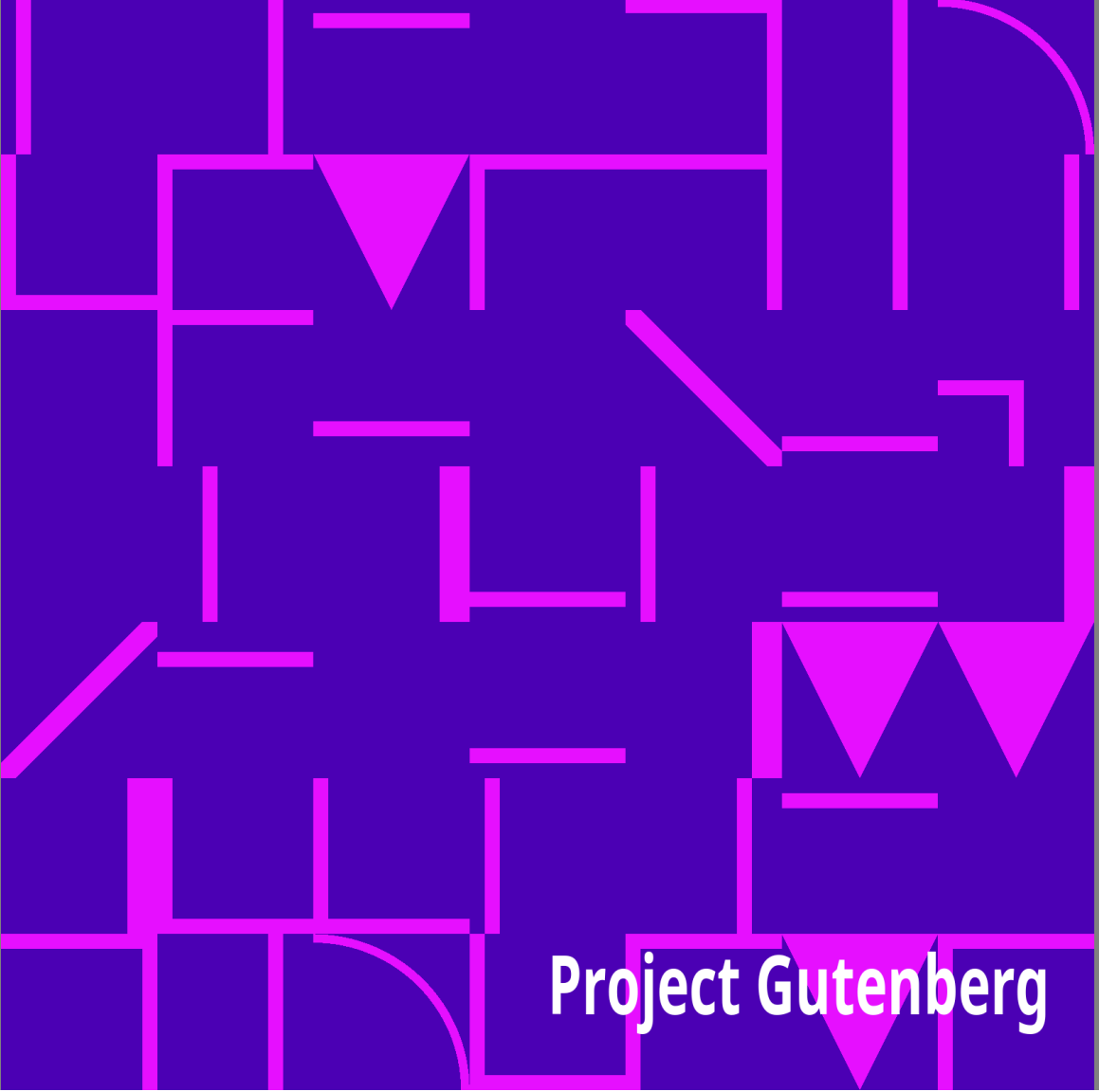
The philosophy of Mr. B*rrtr*nd R*ss*ll

Philip E. B. Jourdain

Project Gutenberg

The philosophy of Mr. B*rrtr*nd R*ss*ll

Philip E. B. Jourdain



Project Gutenberg

The Project Gutenberg eBook of The philosophy of Mr. B*rtr*nd R*ss*ll

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The philosophy of Mr. B*rtr*nd R*ss*ll

Editor: Philip E. B. Jourdain

Release date: December 28, 2011 [eBook #38430]

Language: English

Other information and formats: www.gutenberg.org/ebooks/38430

Credits: Produced by Adrian Mastronardi and the Online Distributed Proofreading Team at <http://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive/Canadian Libraries)

*** START OF THE PROJECT GUTENBERG EBOOK THE
PHILOSOPHY OF MR. B*RTR*ND R*SS*LL ***

THE PHILOSOPHY OF MR. B*RT*ND R*SS*LL

WITH AN APPENDIX OF LEADING
PASSAGES FROM CERTAIN OTHER WORKS

EDITED BY
PHILIP E. B. JOURDAIN



LONDON: GEORGE ALLEN & UNWIN LTD.
RUSKIN HOUSE 40 MUSEUM STREET, W.C. 1
CHICAGO: THE OPEN COURT PUBLISHING CO.

First published in 1918

(All rights reserved)

EDITOR'S NOTE

When Mr. B*rr*nd R*ss*ll, following the advice of Mr. W*ll**m J*m*s, again “got into touch with reality” and in July 1911 was torn to pieces by Anti-Suffragists, many of whom were political opponents of Mr. R*ss*ll and held strong views on the Necessity of Protection of Trade and person, a manuscript which was almost ready for the press was fortunately saved from the flames on the occasion when a body of eager champions of the Sacredness of Personal Property burnt the late Mr. R*ss*ll’s house. This manuscript, together with some further fragments found in the late Mr. R*ss*ll’s own interleaved copy of his *Prayer-Book of Free Man’s Worship*, which was fortunately rescued with a few of the great author’s other belongings, was first given to the world in the *Monist* for October 1911 and January 1916, and has here been arranged and completed by some other hitherto undecipherable manuscripts. The title of the above-mentioned *Prayer-Book*, it may perhaps be mentioned, was apparently suggested to Mr. R*ss*ll by that of the Essay on “The Free Man’s Worship” in the *Philosophical Essays* (London, 1910, pp. 59-70^[1]) of Mr. R*ss*ll’s distinguished contemporary, Mr. Bertrand Russell, from whom much of Mr. R*ss*ll’s philosophy was derived. And, indeed, the influence of Mr. Russell extended even beyond philosophical views to arrangement and literary style. The method of arrangement of the present work seems to have been borrowed from Mr. Russell’s *Philosophy of Leibniz* of 1900; in the selection of subjects dealt with, Mr. R*ss*ll seems to have been guided by Mr. Russell’s *Principles of Mathematics* of 1903; while Mr. R*ss*ll’s literary style fortunately reminds us more of Mr. Russell’s later clear and charming subtleties than his earlier brilliant and no less subtle obscurities. But, on the other hand, some important points of Mr. Russell’s doctrine, which first appeared in books published after Mr. R*ss*ll’s death, were anticipated in Mr. R*ss*ll’s notes, and these anticipations, so interesting for future historians of philosophy, have been provided by the editor with references to the later works of Mr. Russell. All editorial notes are enclosed in square brackets, to indicate that they were not written by the late Mr. R*ss*ll.

At the present time we have come to take a calm view of the question so much debated seven years ago as to the legitimacy of logical arguments in political discussions. No longer, fortunately, can that intense feeling be roused which then found expression in the famous cry, “Justice—right or wrong,” and which played such a large part in the politics of that time. Thus it will not be out of place in this unimpassioned record of some of the truths and errors in the world to refer briefly to Mr. R*ss*ll’s short and stormy career. Before he was torn to pieces, he had been forbidden to lecture on philosophy or mathematics by some well-intentioned advocates of freedom in speech who thought that the cause of freedom might be endangered by allowing Mr. R*ss*ll to speak freely on points of logic, on the grounds, apparently, that logic is both harmful and unnecessary and might be applied to politics unless strong measures were taken for its suppression. On much the same grounds, his liberty was taken from him by those who remarked that, if necessary, they would die in defence of the sacred principle of liberty; and it was in prison that the greater part of the present work was written. Shortly after his liberation, which, like all actions of public bodies, was brought about by the combined honour and interests of those in authority, occurred his lamentable death to which we have referred above.

Mr. R*ss*ll maintained that the chief use of “implication” in politics is to draw conclusions, which are thought to be true, and which are consequently false, from identical propositions, and we can see these views expressed in Chapters III and XIX of the present work. These chapters were apparently written before the Government, in the spring of 1910, arrived at the famous secret decision that only “certain implications” are permitted in discussion. Naturally the secret decision gave rise to much speculation among logicians as to which kinds of implication were barred, and Mr. R*ss*ll and Mr. Bertrand Russell had many arguments on the subject, which naturally could not be published at the time. However, after Mr. R*ss*ll’s death, successive prosecutions which were made by the Government at last made it quite clear that the opinion held by Mr. R*ss*ll was the correct one. There had been numerous prosecutions of people who, from true but not identical premisses, had deduced true conclusions, so that the possible legitimate forms of “implication” were reduced. Further, the other doubtful cases were cleared up in course of time by the prosecution of (1) members of the Aristotelian Society for deducing true conclusions from false premisses; (2)

members of the *Mind* Association for deducing false conclusions from false premisses; and also by the attempted prosecution of an eminent lady for deducing true conclusions from identities. Fortunately this lady was able to defend herself successfully by pleading that one eminent philosopher believed them to be true—which, of course, means that the conclusions are false. Thus appeared the true nature of legitimate political arguments.

[\[1\]](#) [This Essay is also reprinted in Mr. Russell's *Mysticism and Logic*, London and New York, 1918, pp. 46-57.—ED.]

“Even a joke should have some meaning....”

(The Red Queen, *T. L. G.*, p. 105).

CONTENTS

	PAGE
EDITOR'S NOTE	<u>3</u>
ABBREVIATIONS	<u>9</u>
CHAPTER	
I. THE INDEFINABLES OF LOGIC	<u>11</u>
II. OBJECTIVE VALIDITY OF THE "LAWS OF THOUGHT"	<u>15</u>
III. IDENTITY	<u>16</u>
IV. IDENTITY OF CLASSES	<u>18</u>
V. ETHICAL APPLICATIONS OF THE LAW OF IDENTITY	<u>19</u>
VI. THE LAW OF CONTRADICTION IN MODERN LOGIC	<u>21</u>
VII. SYMBOLISM AND MEANING	<u>22</u>
VIII. NOMINALISM	<u>24</u>
IX. AMBIGUITY AND SYMBOLIC LOGIC	<u>26</u>
X. LOGICAL ADDITION AND THE UTILITY OF SYMBOLISM	<u>27</u>
XI. CRITICISM	<u>29</u>
XII. HISTORICAL CRITICISM	<u>30</u>
XIII. IS THE MIND IN THE HEAD?	<u>31</u>
XIV. THE PRAGMATIST THEORY OF TRUTH	<u>32</u>
XV. ASSERTION	<u>34</u>
XVI. THE COMMUTATIVE LAW	<u>35</u>
XVII. UNIVERSAL AND PARTICULAR PROPOSITIONS	<u>36</u>
XVIII. DENIAL OF GENERALITY AND GENERALITY OF DENIAL	<u>37</u>
XIX. IMPLICATION	<u>39</u>
XX. DIGNITY	<u>43</u>
XXI. THE SYNTHETIC NATURE OF DEDUCTION	<u>45</u>
XXII. THE MORTALITY OF SOCRATES	<u>48</u>
XXIII. DENOTING	<u>53</u>
XXIV. THE	<u>54</u>
XXV. NON-ENTITY	<u>56</u>

XXVI. IS	58
XXVII. AND AND OR	59
XXVIII. THE CONVERSION OF RELATIONS	60
XXIX. PREVIOUS PHILOSOPHICAL THEORIES OF MATHEMATICS	61
XXX. FINITE AND INFINITE	63
XXXI. THE MATHEMATICAL ATTAINMENTS OF TRISTRAM SHANDY	64
XXXII. THE HARDSHIPS OF A MAN WITH AN UNLIMITED INCOME	66
XXXIII. THE RELATIONS OF MAGNITUDE OF CARDINAL NUMBERS	69
XXXIV. THE UNKNOWABLE	70
XXXV. MR. SPENCER, THE ATHANASIAN CREED, AND THE ARTICLES	73
XXXVI. THE HUMOUR OF MATHEMATICIANS	74
XXXVII. THE PARADOXES OF LOGIC	75
XXXVIII. MODERN LOGIC AND SOME PHILOSOPHICAL ARGUMENTS	79
XXXIX. THE HIERARCHY OF JOKES	81
XL. THE EVIDENCE OF GEOMETRICAL PROPOSITIONS	83
XLI. ABSOLUTE AND RELATIVE POSITION	84
XLII. LAUGHTER	86
XLIII. “GEDANKENEXPERIMENTE” AND EVOLUTIONARY ETHICS	88
APPENDIXES	89

ABBREVIATIONS

- Lewis Carroll: *Alice's Adventures in Wonderland*, London, 1908.
- A. A. [This book was first published much earlier, but this was the
W. edition used by Mr. R*ss*ll. The same applies to *H. S.* and *T. L. G.*]
- A. C. John Henry Blunt (ed. by): *The Annotated Book of Common
P. Prayer*, London, new edition, 1888.
- A. d. Ernst Schröder: *Vorlesungen über die Algebra der Logik*, Leipzig,
L. vol. i., 1890; vol. ii. (two parts), 1891 and 1905; vol. iii.: *Algebra und Logik der Relative*, 1895.
- E. N. Richard Dedekind: *Essays on the Theory of Numbers*, Chicago and London, 1901.
- E. L. William Stanley Jevons: *Elementary Lessons in Logic, Deductive
L. and Inductive. With copious Questions and Examples, and a Vocabulary of Logical Terms*, London, 24th ed., 1907 [first published in 1870].
- E. u. I. Ernst Mach: *Erkenntnis und Irrtum: Skizzen zur Psychologie der
Forschung*, Leipzig, 1906.
- F. L. Augustus De Morgan: *Formal Logic: or The Calculus of
Inference, Necessary and Probable*, London, 1847.
- Fm. L. John Neville Keynes: *Studies and Exercises in Formal Logic*, 4th ed., London, 1906.
- Gg. Gottlob Frege: *Grundgesetze der Arithmetik begriffsschriftlich
abgeleitet*, Jena, vol. i., 1893; vol. ii., 1903.
- Gl. Gottlob Frege: *Die Grundlagen der Arithmetik: eine logisch-
mathematische Untersuchung über den Begriff der Zahl*, Breslau, 1884.
- G. u. G. Heymans: *Die Gesetze und Elemente des wissenschaftlichen
E. Denkens*, Leiden, vol. i., 1890; vol. ii., 1894.

- H. J. *The Hibbert Journal: a Quarterly Review of Religion, Theology and Philosophy*, London and New York.
- H. S. Lewis Carroll: *The Hunting of the Snark: an Agony in Eight Fits*, London, 1911.
- M. *The Monist: a Quarterly Magazine Devoted to Science and Philosophy*, Chicago and London.
- Md. *Mind: a Quarterly Review of Psychology and Philosophy*, London and New York.
- Pa. Alfred North Whitehead and Bertrand Russell: *Principia*
Ma. *Mathematica*, vol. i., Cambridge, 1910. [Other volumes were published in 1912 and 1913.]
- P. E. Bertrand Russell: *Philosophical Essays*, London and New York, 1910.
- Ph. L. Bertrand Russell: *A Critical Exposition of the Philosophy of Leibniz, with an Appendix of Leading Passages*, Cambridge, 1900.
- P. M. Bertrand Russell: *The Principles of Mathematics*, vol. i., Cambridge, 1903.
- R. M. *Revue de Métaphysique et de Morale*, Paris.
M.
- S. B. Lewis Carroll: *Sylvie and Bruno*, London, 1889.
- S. L. John Venn: *Symbolic Logic*, London, 1881; 2nd ed., 1894.
- S. o. S. William Stanley Jevons: *The Substitution of Similars, the True Principle of Reasoning derived from a Modification of Aristotle's Dictum*, London, 1869.
- T. L. Lewis Carroll: *Through the Looking-Glass, and what Alice found*
G. *there*, London, 1911.
- Z. S. Gottlob Frege: *Ueber die Zahlen des Herrn H. Schubert*, Jena, 1899.
-

CHAPTER I

THE INDEFINABLES OF LOGIC

The view that the fundamental principles of logic consist solely of the law of identity was held by Leibniz,^[2] Drobisch, Uberweg,^[3] and Tweedledee. Tweedledee, it may be remembered,^[4] remarked that certain identities “are” logic. Now, there is some doubt as to whether he, like Jevons,^[5] understood “are” to mean what mathematicians mean by “=,” or, like Schröder^[6] and most logicians, to have the same meaning as the relation of subsumption. The first alternative alone would justify our contention; and we may, I think, conclude from an opposition to authority that may have been indicated by Tweedledee’s frequent use of the word “contrariwise” that he did not follow the majority of logicians, but held, like Jevons,^[7] the mistaken^[8] view that the quantification of the predicate is relevant to symbolic logic.

It may be mentioned, by the way, that it is probable that Humpty-Dumpty’s “is” is the “is” of identity. In fact, it is not unlikely that Humpty-Dumpty was a Hegelian; for, although his ability for clear explanation may seem to militate against this, yet his inability to understand mathematics,^[9] together with his synthesis of a cravat and a belt, which usually serve different purposes,^[10] and his proclivity towards riddles seem to make out a good case for those who hold that he was in fact a Hegelian. Indeed, riddles are very closely allied to puns, and it was upon a pun, consisting of the confusion of the “is” of predication with the “is” of identity—so that, for example, “Socrates” was identified with “mortal” and more generally the particular with the universal—that Hegel’s system of philosophy was founded.^[11] But the question of Humpty-Dumpty’s philosophical opinions must be left for final verification to the historians of philosophy: here I am only concerned with an *a priori* logical construction of what his views might have been if they formed a consistent whole.^[12]

If the principle of identity were indeed the sole principle of logic, the principles of logic could hardly be said to be, as in fact they are, a body of

propositions whose consistency it is impossible to prove.^[13] This characteristic is important and one of the marks of the greatest possible security. For example, while a great achievement of late years has been to prove the consistency of the principles of arithmetic, a science which is unreservedly accepted except by some empiricists,^[14] it can be proved formally that one foundation of arithmetic is shattered.^[15] It is true that, quite lately, it has been shown that this conclusion may be avoided, and, by a re-moulding of logic, we can draw instead the paradoxical conclusion that the opinions held by common-sense for so many years are, in part, justified. But it is quite certain that, with the principles of logic, no such proof of consistency, and no such paradoxical result of further investigations is to be feared.

Still, this re-moulding has had the result of bringing logic into a fuller agreement with common-sense than might be expected. There were only two alternatives: if we chose principles in accordance with common-sense, we arrived at conclusions which shocked common-sense; by starting with paradoxical principles, we arrived at ordinary conclusions. Like the White Knight, we have dyed our whiskers an unusual colour and then hidden them.^[16]

The quaint name of “Laws of Thought,” which is often applied to the principles of Logic, has given rise to confusion in two ways: in the first place, the “Laws,” unlike other laws, cannot be broken, even in thought; and, in the second place, people think that the “Laws” have something to do with holding for the operations of their minds, just as laws of nature hold for events in the world around us.^[17] But that the laws are not psychological laws follows from the facts that a thing may be true even if nobody believes it, and something else may be false if everybody believes it. Such, it may be remarked, is usually the case.

Perhaps the most frequent instance of the assumption that the laws of logic are mental is the treatment of an identity as if its validity were an affair of our permission. Some people suggest to others that they should “let bygones be bygones.” Another important piece of evidence that the truth of propositions has nothing to do with mind is given by the phrase “it is morally certain that such-and-such a proposition is true.” Now, in the first place, morality, curiously enough, seems to be closely associated with

mental acts: we have professorships and lectureships of, and examinations in, “mental and moral philosophy.” In the second place, it is plain that a “morally certain” proposition is a highly doubtful one. Thus it is as vain to expect any information about our minds from a study of the “Laws of Thought” as it would be to expect a description of a certain social event from Miss E. E. C. Jones’s book *An Introduction to General Logic*.

Fortunately, the principles or laws of Logic are not a matter of philosophical discussion. Idealists like Tweedledum and Tweedledee, and even practical idealists like the White Knight, explicitly accept laws like the law of identity and the excluded middle.^[18] In fact, throughout all logic and mathematics, the existence of the human or any other mind is totally irrelevant; mental processes are studied by means of logic, but the subject-matter of logic does not presuppose mental processes, and would be equally true if there were no mental processes. It is true that, in that case, we should not know logic; but our knowledge must not be confounded with the truths which we know.^[19] An apple is not confused with the eating of it except by savages, idealists, and people who are too hungry to think.

[2] Russell, *Ph. L.*, pp. 17, 19, 207-8.

[3] Schröder, *A. d. L.*, i. p. 4.

[4] See [Appendix A](#). This Appendix also illustrates the importance attached to the Principle of Identity by the Professor and Bruno.

[5] *S. o. S.*, pp. 9-15.

[6] *A. d. L.*, i. p. 132.

[7] Cf., besides the reference in the last note but one, *E. L. L.*, pp. 183, 191. “Contrariwise,” it may be remarked, is not a term used in traditional logic.

[8] *S. L.*, 1881, pp. 173-5, 324-5; 1894, pp. 194-6.

[9] Cf. [Appendix C](#), and William Robertson Smith, “Hegel and the Metaphysics of the Fluxional Calculus,” *Trans. Roy. Soc., Edinb.*, vol. xxv., 1869, pp. 491-511.

[10] See [Appendix B](#).

[11] [This is a remarkable anticipation of the note on pp. 39-40 of Mr. Russell’s book, published about three years after the death of Mr. R*ss*ll, and

entitled *Our Knowledge of the External World as a Field for Scientific Method in Philosophy*, Chicago and London, 1914.—ED.]

[12] Cf. *Ph. L.*, pp. v.-vi. 3.

[13] Cf. Pieri, *R. M. M.*, March 1906, p. 199.

[14] As a type of these, Humpty-Dumpty, with his inability to admit anything not empirically given and his lack of comprehension of pure mathematics, may be taken (see [Appendix C](#)). In his (correct) thesis that definitions are nominal, too, Humpty-Dumpty reminds one of J. S. Mill (see [Appendix D](#)).

[15] See Frege, *Gg.*, ii. p. 253.

[16] See [Appendix E](#).

[17] See Frege, *Gg.*, i. p. 15.

[18] See the above references and also [Appendix F](#).

[19] Cf. B. Russell, *H. J.*, July 1904, p. 812.

CHAPTER II

OBJECTIVE VALIDITY OF THE “LAWS OF THOUGHT”

I once inquired of a maid-servant whether her mistress was at home. She replied, in a doubtful fashion, that she *thought* that her mistress was in unless she was out. I concluded that the maid was uncertain as to the objective validity of the law of excluded middle, and remarked that to her mistress. But since I used the phrase “laws of thought,” the mistress perhaps supposed that a “law of thought” has something to do with thinking and seemed to imagine that I wished to impute to the maid some moral defect of an unimportant nature. Thus she remonstrated with me in an amused way, since she probably imagined that I meant to find fault with the maid’s capacity for thinking.

CHAPTER III

IDENTITY

In the first chapter we have noticed the opinion that identities are fundamental to all logic. We will now consider some other views of the value of identities.

Identities are frequently used in common life by people who seem to imagine that they can draw important conclusions respecting conduct or matters of fact from them. I have heard of a man who gained the double reputation of being a philosopher and a fatalist by the repeated enunciation of the identity “Whatever will be, will be”; and the Italian equivalent of this makes up an appreciable part of one of Mr. Robert Hichens’ novels. Further, the identity “Life is Life” has not only been often accepted as an explanation for a particular way of living but has even been considered by an authoress who calls herself “Zack” to be an appropriate title for a novel; while “Business is Business” is frequently thought to provide an excuse for dishonesty in trading, for which purpose it is plainly inadequate.

Another example is given by a poem of Mr. Kipling, where he seems to assert that “East is East” and “West is West” imply that “never the twain shall meet.” The conclusion, now, is false; for, since the world is round—as geography books still maintain by arguments which strike every intelligent child as invalid^[20]—what is called the “West” does, in fact, merge into the “East.” Even if we are to take the statement metaphorically, it is still untrue, as the Japanese nation has shown.

The law-courts are often rightly blamed for being strenuous opponents of the spread of modern logic: the frequent misuse of *and*, *or*, *the*, and *provided that* in them is notorious. But the fault seems partly to lie in the uncomplicated nature of the logical problems which are dealt with in them. Thus it is no uncommon thing for somebody to appear there who is unable to establish his own identity, or for A to assert that B was “not himself” when he made a will leaving his money to C.

The chief use of identities is in implication. Since, in logic, we so understand *implication*^[21] that any true proposition implies and is implied by any other true proposition; if one is convinced of the truth of the proposition Q, it is advisable to choose one or more identities P, whose truth is undoubted, and say that P implies Q. Thus, Mr. Austen Chamberlain, according to *The Times* of March 27, 1909, professed to deduce the conclusion that it is not right that women should have votes from the premisses that “man is man” and “woman is woman.” This method requires that one should have made up one’s mind about the conclusion before discovering the premisses—by what, no doubt, Jevons would call an “inverse or inductive method.” Thus the method is of use only in speeches and in giving good advice.

Mr. Austen Chamberlain afterwards rather destroyed one’s belief in the truth of his premisses by putting limits to the validity of the principle of identity. In the course of the Debate on the Budget of 1909, he maintained, against Mr. Lloyd George, that a joke was a joke except when it was an untruth: Mr. Lloyd George, apparently, being of the plausible opinion that a joke is a joke under all circumstances.

^[20] The argument about the hull of a ship disappearing first is not convincing, since it would equally well prove that the surface of the earth was, for example, corrugated on a large scale. If the common-sense of the reader were supposed to dismiss the possibility of water clinging to such corrugations, it might equally be supposed to dismiss the possibility of water clinging to a spherical earth. Traditional geography books, no doubt, gave rise to the opinions held by Lady Blount and the Zetetic Society.

^[21] The subject of Implication will be further considered in Chapter XIX.

CHAPTER IV

IDENTITY OF CLASSES

I once heard of a meritorious lady who was extremely conventional; on the slender grounds of carefully acquired habits of preferring the word “woman” to the word “lady” and of going to the post-office without a hat, imagined that she was unconventional and altogether a remarkable person; and who once remarked with great satisfaction that she was a “very queer person,” and that nothing shocked her “except, of course, bad form.”

Thus, she asserted that all the things which shocked her were actions in bad form; and she would undoubtedly agree, though she did not actually state it, that all the things which were done in bad form would shock her. Consequently she asserted that the class of things which shocked her was the class of actions in bad form. Consequently the statement of this lady that some or all of the actions done in bad form shocked her is an identical proposition of the form “nothing shocks me, except, of course, the things which do, in fact, shock me”; and this statement the lady certainly did not intend to make.

This excellent lady, had she but known it, was logically justified in making any statement whatever about her unconventionality. For the class of her unconventional actions was the null class. Thus she might logically have made inconsistent statements about this class of actions. As a matter of fact she did make inconsistent statements, but unfortunately she justified them by stating that, “It is the privilege of woman to be inconsistent.” She was one of those persons who say things like that.

CHAPTER V

ETHICAL APPLICATIONS OF THE LAW OF IDENTITY

It may be remembered that Mr. Podsnap remarked, with sadness tempered by satisfaction, that he regretted to say that “Foreign nations do as they do do.” Besides aiding the comforting expression of moral disapproval, the law of identity has yet another useful purpose in practical ethics: It serves the welcome purpose of providing an excuse for infractions of the moral law. There was once a man who treated his wife badly, was unfaithful to her, was dishonest in business, and was not particular in his use of language; and yet his life on earth was described in the lines:

This man maintained a wife’s a wife,
Men are as they are made,
Business is business, life is life;
And called a spade a spade.

One of the objects of Dr. G. E. Moore’s *Principia Ethica*^[22] was to argue that the word “good” means simply good, and not pleasant or anything else. Appropriately enough, this book bore on its title-page the quotation from the preface to the *Sermons*, published in 1726, of Bishop Joseph Butler, the author of the *Analogy*: “Everything is what it is and not another thing.”

But another famous Butler—Samuel Butler, the author of *Hudibras*—went farther than this, and maintained that identities were the highest attainment of metaphysics itself. At the beginning of the first Canto of *Hudibras*, in the description of Hudibras himself, Butler wrote:

He knew what’s what, and that’s as high
As metaphysic wit can fly.

I once conducted what I imagined to be an æsthetic investigation for the purpose of discovery, by the continual use of the word “Why?”^[23] the grounds upon which certain people choose to put milk into a tea-cup before the tea. I was surprised to discover that it was an ethical, and not an æsthetic

problem; for I soon elicited the fact that it was done because it was “right.” A continuance of my patient questioning elicited further evidence of the fundamental character of the principle of identity in ethics; for it was right, I learned, because “right is right.”

It appears that some people unconsciously think that the principle of identity is the foundation, in certain religions, of the reasons which can be alleged for moral conduct, and are surprised when this fact is pointed out to them. The late Sir Leslie Stephen, when travelling by railway, fell into conversation with an officer of the Salvation Army, who tried hard to convert him. Failing in this laudable endeavour, the Salvationist at last remarked: “But if you aren’t saved, you can’t go to heaven!” “That, my friend,” replied Stephen, “is an identical proposition.”

[22] Cambridge, 1903.

[23] Cf. *P. E.*, p. 2.

CHAPTER VI

THE LAW OF CONTRADICTION IN MODERN LOGIC

Considering the important place assigned by philosophers and logicians to the law of contradiction, the remark will naturally be resented by many of the older schools of philosophy, and especially by Kantians, that “in spite of its fame we have found few occasions for its use.”^[24] Also in modern times, Benedetto Croce, an opponent of both traditional logic and mathematical logic, began the preface of the book of 1908 on Logic^[25] by saying that that volume “is and is not” a certain memoir of his which had been published in 1905.

^[24] *Pa. Ma.*, p. 116.

^[25] [English translation of the third Italian edition by Douglas Ainslie, under the title: *Logic as the Science of the Pure Concept*, London 1917.—ED.]

CHAPTER VII

SYMBOLISM AND MEANING

When people write down any statement such as “The curfew tolls the knell of parting day,”^[26] which we will call “C” for shortness, what they mean is not “C” but the *meaning* of “C”; and not “the meaning of ‘C’” but the *meaning* of “the meaning of ‘C’.” And so on, *ad infinitum*. Thus, in writing or in speech, we always fail to state the meaning of any proposition whatever. Sometimes, indeed, we succeed in *conveying* it; but there is danger in too great a disregard of statement and preoccupation with conveyance of meaning. Thus many mathematicians have been so anxious to convey to us a perfectly distinct and unmetaphysical concept of number that they have stripped away from it everything that they considered unessential (like its logical nature) and have finally delivered it to us as a mere *sign*. By the labours of Helmholtz, Kronecker, Heine, Stolz, Thomae, Pringsheim, and Schubert, many people were persuaded that, when they said “‘2’ is a number” they were speaking the truth, and hold that “Paris” is a town containing the letter “P.” When Frege pointed out^[27] this difficulty he was almost universally denounced in Germany as “*spitzfindig*.” In fact, Germans seem to have been influenced perhaps by that great contemner of “*Spitzfindigkeit*,” Kant, to reject the White Knight’s^[28] distinctions between words and their denotations and to regard subtlety with disfavour to such a degree that their only mathematical logician except Frege, namely Schröder—the least subtle of mortals, by the way—seems to have been filled with such fear of being thought subtle, that he made his books so prolix that nobody has read them.

Another term which, as we shall see when discussing the paradoxes of logic, mathematicians are accustomed to apply to thought which is more exact than any to which they are accustomed is “scholastic.”^[29] By this, I suppose, they mean that the pursuits of certain acute people of the Middle Ages are unimportant in contrast with the great achievements of modern thought, as exemplified by a method of making plausible guesses known as

induction,^[30] the bicycle, and the gramophone—all of them instruments of doubtful merit.

[26] Cf. *Md*, N. S., vol. xiv., October 1905, p. 486.

[27] In *Z. S.*, for example.

[28] See [Appendix G](#).

[29] Cf. Chapter XXXVII below.

[30] Cf. *P. M.*, p. 11, note.

CHAPTER VIII

NOMINALISM

De Morgan^[31] said that, “if all mankind had spoken one language, we cannot doubt that there would have been a powerful, perhaps universal, school of philosophers who would have believed in the inherent connexion between names and things; who would have taken the sound *man* to be the mode of agitating the air which is essentially communicative of the ideas of reason, cookery, bipedality, etc.... ‘The French,’ said the sailor, ‘call a cabbage a *shoe*; the fools! Why can’t they call it a cabbage, when they must know it is one?’”

One of the chief differences between logicians and men of letters is that the latter mean many different things by one word, whereas the former do not—at least nowadays. Most mathematicians belong to the class of men of letters.

I once had a manservant who told me on a certain occasion that he “never thought a word about it.” I was doubtful whether to class him with such eminent mathematicians as are mentioned in the last chapter, or as a supporter of Max Müller’s theory of the identity of thought and language. However, since the man was very untruthful, and he told me that he meant what he said and said what he meant,^[32] the conclusion is probably correct that he really believed that the meanings of his words were not the words themselves. Thus I think it most probable that my manservant had been a mathematician but had escaped by the aid of logic.

As regards his remark that he meant what he said and said what he meant, he plainly wished to pride himself on certain virtues which he did not possess, and was not indifferent to applause, which, however, was never evoked. The virtues, if so they be, and the applause were withheld for other reasons than that the above statements are either nonsensical or false. Suppose that “I say what I mean” expresses a truth. What I say (or write) is always a symbol—words (or marks); and what I mean by the symbol is the meaning of the symbol and not the symbol itself. So the remark cannot

express a truth, any more than the name “Wellington” won the battle of Waterloo.

[31] *F. L.*, pp. 246-7.

[32] The Hatter (see [Appendix H](#)) pointed out that there is a difference between these two assertions. Thus, he clearly showed that he was a nominalist, and philosophically opposed to the March Hare who had recommended Alice to say what she meant.

CHAPTER IX

AMBIGUITY AND SYMBOLIC LOGIC

The universal use of some system of Symbolic Logic would not only enable everybody easily to deal with exceedingly complicated arguments, but would prevent ambiguous arguments. In denying the indispensability of Symbolic Logic in the former state of things, Keynes^[33] is probably alone, against the need strongly felt by Alice when speaking to the Duchess,^[34] and most modern logicians. It may be noticed that the Duchess is more consistent than Keynes, for Keynes really uses the signs for logical multiplication and addition of Boole and Venn under the different shapes of the words “and” and “or.”

As regards ambiguity, a translation of *Hymns Ancient and Modern* into, say, Peanesque, would prevent the puzzle of childhood as to whether the “his” in

And Satan trembles when he sees
The weakest saint upon his knees

refers to the saint’s knees or Satan’s.

[33] In his *Fm. L.*

[34] See [Appendix I](#).

CHAPTER X

LOGICAL ADDITION AND THE UTILITY OF SYMBOLISM

Frequently ordinary language contains subtle psychological implications which cannot be translated into symbolic logic except at great length. Thus if a man (say Mr. Jones) wishes to speak collectively of himself and his wife, the order of mentioning the terms in the class considered and the names applied to these terms are, logically speaking, irrelevant. And yet more or less definite information is given about Mr. Jones according as he talks to his friends of:

- (1) Mrs. Jones and I,
- (2) I (or me) and my wife (or missus),
- (3) My wife and I,
- or (4) I (or me) and Mrs. Jones.

In case (1) one is probably correct in placing Mr. Jones among the clergy or the small professional men who make up the bulk of the middle-class; in case (2) one would conclude that Mr. Jones belonged to the lower middle-class; the form (3) would be used by Mr. Jones if he were a member of the upper, upper middle, or lower class; while form (4) is only used by retired shopkeepers of the lower middle-class, of which a male member usually combines belief in the supremacy of man with belief in the dignity of his wife as well as himself. A further complication is introduced if a wife is referred to as “the wife.”^[35] Cases (2) and (3) then each give rise to one more case. Cases (1) and (4) do not, since nobody has hitherto referred to his wife as “the Mrs. Jones”—at least without a qualifying adjective before the “Mrs.”

On the other hand, certain descriptive phrases and certain propositions can be expressed more shortly and more accurately by means of symbolic logic. Let us consider the proposition “No man marries his deceased wife’s sister.” If we assume, as a first approximation, that all marriages are fertile

and that all children are legitimate, then, with only four primitive ideas: the relation of parent to child (P) and the three classes of males, females, and dead people, we can define “wife” (a female who has the relation formed by taking the relative product of P and $\check{P}$ ^[36] to a male), “sister,” “deceased wife,” and “deceased wife’s sister” in terms of these ideas and of the fundamental notions of logic. Then the proposition “No man marries his deceased wife’s sister” can be expressed unambiguously by about twenty-nine simple signs on paper, whereas, in words, the unasserted statement consists of no less than thirty-four letters. Although, legally speaking, we should have to adopt somewhat different definitions and possibly increase the complications of our proposition, it must be remembered that, on the other hand, we always reduce the number of symbols in any proposition by increasing the number of definitions in the preliminaries to it.

But the utility of symbolic logic should not be estimated by the brevity with which propositions may sometimes be expressed by its means. Logical simplicity, in fact, can very often only be obtained by apparently complicated statements. For example, the logical interpretation of “The father of Charles II was executed” is, “It is not always false of x that x begat Charles II, and that x was executed and that ‘if y begat Charles II, y is identical with x ’ is always true of y .”^[37] From the point of view of logic, we may say that the apparently simple is most often very complicated, and, even if it is not so, symbolism will make it seem so,^[38] and thus draw attention to what might otherwise easily be overlooked.

[35] Cf. Chapter XXIV below.

[36] C. S. Peirce’s notation for the relation “converse of P.”

[37] Russell, *Md.*, N. S., vol. xiv., October 1905, p. 482.

[38] Russell, *International Monthly*, vol. iv., 1901, pp. 85-6; cf. *M.*, vol. xxii., 1912, p. 153. [This essay is reprinted in *Mysticism and Logic*, London and New York, 1918, pp. 74-96.—ED.]

CHAPTER XI

CRITICISM

Those people who think that it is more godlike to seem to turn water into wine than to seem to turn wine into water surprise me. I cannot imagine an intolerable critic. It seems to me that, if A resents B's criticism in trying to put his (A's) discovery in the right or wrong place, A acts as if he thought he had some private property in truth. The White Queen seems to have shared the popular misconception as to the nature of criticism. ^[39]

^[39] See [Appendix J](#).

CHAPTER XII

HISTORICAL CRITICISM

From a problem in Diophantus's *Arithmetic* about the price of some wine it would seem that the wine was of poor quality, and Paul Tannery has suggested that the prices mentioned for such a wine are higher than were usual until after the end of the second century. He therefore rejected the view which was formerly held that Diophantus lived in that century.^[40]

The same method applied to a problem given by the ancient Hindu algebraist Brahmagupta, who lived in the seventh century after Christ, might result in placing Brahmagupta in prehistoric times. This is the problem:^[41] “Two apes lived at the top of a cliff of height h , whose base was distant mh from a neighbouring village. One descended the cliff and walked to the village, the other flew up a height x and then flew in a straight line to the village. The distance traversed by each was the same. Find x .”

^[40] W. W. Rouse Ball, *A Short Account of the History of Mathematics*, 4th edition, London, 1908, p. 109.

^[41] *Ibid.*, pp. 148-9.

CHAPTER XIII

IS THE MIND IN THE HEAD?

The contrary opinion has been maintained by idealists and a certain election agent with whom I once had to deal and who remarked that something slipped his mind and then went out of his head altogether. At some period, then, a remembrance was in his head and out of his mind; his mind was not, then, wholly within his head. Also, one is sometimes assured that with certain people “out of sight is out of mind.” What is in their minds is therefore in sight, and cannot therefore be inside their heads.

CHAPTER XIV

THE PRAGMATIST THEORY OF TRUTH

The pragmatist theory that “truth” is a belief which works well sometimes conflicts with common-sense and not with logic. It is commonly supposed that it is always better to be sometimes right than to be never right. But this is by no means true. For example, consider the case of a watch which has stopped; it is exactly right twice every day. A watch, on the other hand, which is always five minutes slow is never exactly right. And yet there can be no question but that a belief in the accuracy of the watch which was never right would, on the whole, produce better results than such a belief in the one which had altogether stopped. The pragmatist would, then, conclude that the watch which was always inaccurate gave truer results than the one which was sometimes accurate. In this conclusion the pragmatist would seem to be correct, and this is an instance of how the false premisses of pragmatism may give rise to true conclusions.

From the text written above the church clock in a certain English village, “Be ye ready, for ye know not the time,” it would be concluded that the clock never stopped for a period as long as twelve hours. For the text is rather a vague symbolical expression of a propositional function which is asserted to be true at all instants. The proposition that a presumably not illiterate and credulous observer of the clock at any definite instant does not know the time implies, then, that the clock is always wrong. Now, if the clock stopped for twelve hours, it would be absolutely right at least once. It must be right twice if it were right at the first instant it stopped or the last instant at which it went;^[42] but the second possibility is excluded by hypothesis, and the occurrence of the first possibility—or of the analogous possibility of the stopped clock being right three times in twenty-four hours—does not affect the present question. Hence the clock can never stop for twelve hours.

The pragmatist’s criterion of truth appears to be far more difficult to apply than the Bellman’s,^[43] that what he said three times is true, and to give results just as insecure.

[42] Both cases cannot occur; the question is similar to that arising in the discussion of the mortality of Socrates (see Chapter XXII).

[43] See [Appendix K](#).

CHAPTER XV

ASSERTION

The subject of the present chapter must not be confused with the assertion of ordinary life. Commonly, an unasserted proposition is synonymous with a probably false statement, while an asserted proposition is synonymous with one that is certainly false. But in logic we apply assertion also to true propositions, and, as Lewis Carroll showed in his version of “What the Tortoise said to Achilles,”^[44] usually pass over unconsciously an infinite series of implications in so doing. If p and q are propositions, p is true, and p implies q , then, at first sight, one would think that one might assert q . But, from (A) p is true, and (B) p implies q , we must, in order to deduce (Z) q is true, accept the hypothetical: (C) If A and B are true, Z must be true. And then, in order to deduce Z from A, B, and C, we must accept another hypothetical: (D) If A, B, and C are true, Z must be true; and so on *ad infinitum*. Thus, in deducing Z, we pass over an infinite series of hypotheticals which increase in complexity. Thus we need a new principle to be able to assert q .

Frege was the first logician sharply to distinguish between an asserted proposition, like “A is greater than B,” and one which is merely considered, like “A’s being greater than B,” although an analogous distinction had been made in our common discourse on certain psychological grounds, for long previously. In fact, soon after the invention of speech, the necessity of distinguishing between a considered proposition and an asserted one became evident, on account of the state of things referred to at the beginning of this chapter.

^[44] *Md. N. S.*, vol. iv., 1895, pp. 278-80. Cf. Russell, *P. M.*, p. 35.

CHAPTER XVI

THE COMMUTATIVE LAW

Often the meaning of a sentence tacitly implies that the commutative law does not hold. We are all familiar with the passage in which Macaulay pointed out that, by using the commutative law because of exigencies of metre, Robert Montgomery unintentionally made Creation tremble at the Atheist's nod instead of the Almighty's. This use of the commutative law by writers of verse renders it doubtful whether, in the hymn-line:

The humble poor believe,

we are to understand a statement about the humble poor, or a doubtful maxim as to the attitude of our minds to statements made by the humble poor.

The non-commutativity of English titles offers difficulties to some novelists and Americans who refer to Mary Lady So-and-So as Lady Mary So-and-So, and *vice versa*.

CHAPTER XVII

UNIVERSAL AND PARTICULAR PROPOSITIONS

People who are cynical as to the morality of the English are often unpleasantly surprised to learn that “All trespassers will be prosecuted” does not necessarily imply that “some trespassers will be prosecuted.” The view that universal propositions are non-existential is now generally held: Bradley and Venn seem to have been the first to hold this, while older logicians, such as De Morgan,^[45] considered universal propositions to be existential, like particular ones.

If the Gnat^[46] had been content to affirm his proposition about the means of subsistence of Bread-and-Butter flies, in consequence of their lack of which such flies always die, without pointing out such an insect and thereby proving that the class of them is not null, Alice’s doubt as to the existence of the class in question, even if it were proved to be well founded, would not have affected the validity of the proposition.

This brings us to a great convenience in treating universal propositions as non-existential: we can maintain that all x ’s are y ’s at the same time as that no x ’s are y ’s, if only x is the null-class. Thus, when Mr. MacColl^[47] objected to other symbolic logicians that their premisses imply that all Centaurs are flower-pots, they could reply that their premisses also imply the more usual view that Centaurs are not flower-pots.

^[45] Cf., e.g., *F. L.*, p. 4.

^[46] See [Appendix L](#).

^[47] Cf., e.g., *Md.*, N. S., vol. xiv., July, 1905, pp. 399-400.

CHAPTER XVIII

DENIAL OF GENERALITY AND GENERALITY OF DENIAL

The conclusion of a certain song^[48] about a young man who poisoned his sweetheart with sheep's-head broth, and was frightened to death by a voice exclaiming:

“Where’s that young maid
What you did poison with my head?”

at his bedside, gives rise to difficulties which are readily solved by a symbolism that brings into relief the principle that the denial of a universal and non-existential proposition is a particular and existential one. The conclusion of the song is:

Now all young men, both high and low,
Take warning by this dismal go!
For if he’d never done nobody no wrong,
He might have been here to have heard this song.

It is an obvious error, say Whitehead and Russell,^[49] though one easy to commit, to assume that the cases: (1) all the propositions of a certain class are true; and (2) no proposition of the class is true; are each other’s contradictories. However, in the modification^[50] of Frege’s symbolism which was used by Russell

(1) is $(x). x$,
and (2) is $(x). \text{not } x$;

while the contradictory of (1) is:

$\text{not } (x). x$.

The last line but one of the above verse may, then, be written:

$$(t). \text{not } (x). \text{not not } \phi(x, t),$$

where “ $\phi(x, t)$ ” denotes the unasserted propositional function “the doing wrong to the person x at the instant t .” By means of the principle of double negation we can at once simplify the above expression into:

$$(t). \text{not } (x). \phi(x, t);$$

which can be thus read: “If at every instant of his life there was at least one person x to whom he did no wrong (at that instant).” It is difficult to imagine any one so sunk in iniquity that he would not satisfy this hypothesis. We are forced, then, unless our imagination for evil is to be distrusted, to conclude that any one might have been there to have heard that song. Now this conclusion is plainly false, possibly on physical grounds, and certainly on æsthetic grounds. It may be added, by the way, that it is quite possible that De Morgan was mistaken in his interpretation of the above proposition owing to the fact that he was unacquainted with Frege’s work. In fact, if he had not noticed the fact that *any* two of the “not’s” cannot be cancelled against one another he would have concluded that the interpretation was: “If he had never done any wrong to anybody.”

According as the symbol for “not” comes before the (x) or between the (x) and the ϕ , we have an expression of what Frege called respectively the denial of generality, and the generality of denial. The denial of the generality of a denial is the form of all existential propositions, while the assertion of or denial of generality is the general form of all non-existential or universal propositions.

[48] To which De Morgan drew attention in a letter; see (Mrs.) S. E. De Morgan, *Memoir of Augustus De Morgan*, London, 1882, p. 324.

[49] *Pa. Ma.*, p. 16.

[50] However, here, for the printer’s convenience, we depart from Mr. Russell’s usage so far as to write “not” for a curly minus sign.

CHAPTER XIX

IMPLICATION

A good illustration of the fact that what is called “implication” in logic is such that a false proposition implies any other proposition, true or false, is given by Lewis Carroll’s puzzle of the three barbers.^[51]

Allen, Brown, and Carr keep a barber’s shop together; so that one of them must be in during working hours. Allen has lately had an illness of such a nature that, if Allen is out, Brown must be accompanying him. Further, if Carr is out, then, if Allen is out, Brown must be in for obvious business reasons. The problem is, may Carr ever go out?

Putting p for “Carr is out,” q for “Allen is out” and r for “Brown is out,” we have:

- (1) q implies r ,
- (2) p implies that q implies not- r .

Lewis Carroll supposed that “ q implies r ” and “ q implies not- r ” are inconsistent, and hence that p must be false. But these propositions are not inconsistent, and are, in fact, both true if q is false. The contradictory of “ q implies r ” is “ q does not imply r ” which is not a consequence of “ q implies not- r .” It seems to be true theoretically that, if Mr. X is a Christian, he is not an Atheist, but we cannot conclude from this alone that his being a Christian does not imply that he is an Atheist, unless we assume that the class of Christians is not null. Thus, if p is true, q is false; or, if Carr is out, Allen is in. The odd part of this conclusion is that it is the one which common-sense would have drawn in that particular case.

A distinguished philosopher (M) once thought that the logical use of the word “implication”—any false proposition being said to “imply” any proposition true or false—is absurd, on the grounds that it is ridiculous to suppose that the proposition “2 and 2 make 5” implies the proposition “M is the Pope.” This is a most unfortunate instance, because it so happens that the false proposition that 2 and 2 make 5 can rigorously be proved to imply

that M, or anybody else other than the Pope, is the Pope. For if 2 and 2 make 5, since they also make 4, we would conclude that 5 is equal to 4. Consequently, subtracting 3 from both sides, we conclude that 2 would be equal to 1. But if this were true, since M and the Pope are two, they would be one, and obviously then M would be the Pope.

The principle that the false implies the true has very important applications in political arguments. In fact, it is hard to find a single principle of politics of which false propositions are not the main support.

If p and q are two propositions, and p implies q ; then, if, and only if, q and p are both false or both true, we also have: q implies p . The most important applications of this invertibility were made by the late Samuel Butler^[52] and Mr. G. B. Shaw. A political application may be made as follows: In a country where only those with middling-sized incomes are taxed, conservative and *bourgeois* politicians would still maintain that the proposition “the rich are taxed” implies the proposition “the poor are taxed,” and this implication, which is true because both premiss and conclusion are false, would be quite unnecessarily supported by many false practical arguments. It is equally true that “the poor are taxed” implies that “the rich are taxed.” And this can be proved, in certain cases, on other grounds. For the taxation of the poor would imply, ultimately, that the poor could not afford to pay a little more for the necessities of life than, in strict justice, they ought; and this would mean the cessation of one of the chief means of production of individual wealth.

We also see why a valuable means for the discovery of truth is given by the inversion of platitudinous implications. It may happen that another platitude is the result of inversion; but it is the fate of any true remark, especially if it is easy to remember by reason of a paradoxical form, to become a platitude in course of time. There are rare cases of a platitude remaining unrepeated for so long that, by a converse process, it has become paradoxical. Such, for example, is Plato’s remark that a lie is less important than an error in thought.

Of late years, a method of disguising platitudes as paradoxes has been too extensively used by Mr. G. K. Chesterton. The method is as follows. Take any proposition p which holds of an entity a ; choose p so that it seems

plausible that p also holds of at least two other entities b and c ; call a , b , c , and any others for which p holds or seems to hold, the class A , and p the “A-ness” or “A-ity” of A ; let d be an entity for which p does not hold; and put d among the A ’s when you think that nobody is looking. Then state your paradox: “Some A ’s do not have A-ness.” By further manipulation you can get the proposition “No A ’s have A-ness.” But it is possible to make a very successful *coup* if A is the null-class, which has the advantage that manipulation is unnecessary. Thus, Mr. Chesterton, in his *Orthodoxy* put A for the class of doubters who doubt the possibility of logic, and proved that such agnostics refuted themselves—a conclusion which seems to have pleased many clergymen.

In this way, Mr. Chesterton has been enabled readily to write many books and to maintain, on almost every page, such theses as that simplicity is not simple, heterodoxy is not heterodox, poets are not poetical, and so on; thereby building up the gigantic platitude that Mr. Chesterton is Chestertonian.

In the chapter on Identity we have illustrated the use of a case of the principle that any proposition implies any true proposition. This important principle may be called *the principle of the irrelevant premiss*,^[53] and is of great service in oratory, because it does not matter what the premiss is, true or false. There is a *principle of the irrelevant conclusion*, but, except in law-courts, interruptions of meetings, and family life, this is seldom used, partly because of the limitation involved in the logical impossibility for the conclusion to be false if the premiss be true, but chiefly because the conclusion is more important than the premiss, being usually a matter of prejudice.

Certain modern logicians, such as Frege, have found it necessary so to extend the meaning of implication of q by p that it holds when p is not a proposition at all. Hitherto, politicians, finding that either identical or false propositions are sufficient for their needs, have made no use of this principle; but it is obvious that their stock of arguments would be vastly increased thereby.

Logical implication is often an enemy of dignity and eloquence. De Morgan^[54] relates “a tradition of a Cambridge professor who was once

asked in a mathematical discussion, ‘I suppose you will admit that the whole is greater than its part?’ and who answered, ‘Not I, until I see what use you are going to make of it.’” And the care displayed by cautious mathematicians like Poincaré, Schoenflies, Borel, Hobson, and Baire in abstaining from pushing their arguments to their logical conclusions is probably founded on the unconscious—but no less well-grounded—fear of appearing ridiculous if they dealt with such extreme cases as “the series of all ordinal numbers.”^[55] They are, probably, as unconscious of implication as Gibbon, when he remarked that he always had a copy of Horace in his pocket, and often in his hand, was of the necessary implication of these propositions that his hand was sometimes in his pocket.

[51] *Md.*, N. S., vol. iii., 1894, pp. 436-8. Cf. the discussions by W. E. Johnson (*ibid.*, p. 583) and Russell (*P. M.*, p. 18, note, and *Md.*, N. S., vol. xiv., 1905, pp. 400-1).

[52] The inhabitants of “Erewhon” punished invalids more severely than criminals. In modern times, one frequently hears the statement that crime is a disease; and if so, it is surely false that criminals ought to be punished.

[53] *Irrelevant* in a popular sense; one would not say, speaking loosely, that the fact that Brutus killed Cæsar implies that the sea is salt; and yet this conclusion is implied both by the above premiss, and the premiss that Cæsar killed Brutus. Cf. on such questions Venn, *S. L.*, 2nd ed., pp. 240-4.

[54] *F. L.*, p. 264.

[55] Cf. Chapters XXIX and XXXVII.

CHAPTER XX

DIGNITY

We have seen, at the end of the preceding chapter, that logical implication is often an enemy of dignity. The subject of dignity is not usually considered in treatises on logic, but, as we have remarked, many mathematicians implicitly or explicitly seem to fear either that the dignity of mathematics will be impaired if she follows out conclusions logically, or that only an act of faith can save us from the belief that, if we followed out conclusions logically, we should find out something alarming about the past, present, or future of mathematics.

Thus it seems necessary to inquire rather more closely into the nature of dignity, with a view to the discovery of whether it is, as is commonly supposed, a merit in life and logic.

The chief use of dignity is to veil ignorance. Thus, it is well known that the most dignified people, as a rule, are schoolmasters, and schoolmasters are usually so occupied with teaching that they have no time to learn anything. And because dignity is used to hide ignorance, it is plain that impudence is not always the opposite of dignity, but that dignity is sometimes impudence. Dignity is said to inspire respect; and this may be in part why respect for others is an error of judgment and self-respect is ridiculous.

Self-respect is, of course, self-esteem. William James has remarked that self-esteem depends, not simply upon our success, but upon the ratio of our success to our pretensions, and can therefore be increased by diminishing our pretensions. Thus if a man is successful, but only then, can he be both ambitious and dignified. James also implies that happiness increases with self-esteem. Likeness of thought with one's friends, then, does not make one happy, for otherwise a man who esteemed himself little would be indeed happy. Also if a man is unhappy he could not, from our premisses, by the principles of the syllogism and of contraposition, be dignified—a conclusion which should be fatal to many novelists' heroes.

A reflection on pessimism to which this discussion gives rise is the following: It would appear that a man's self-esteem would be increased by a conviction of the unworthiness of his neighbours. A man, therefore, who thinks that the world and all its inhabitants, except himself, are very bad, should be extremely happy. In fact, the effects would hardly be distinguishable from those of optimism. And optimism, as everybody knows, is a state of mind induced by stupidity.

CHAPTER XXI

THE SYNTHETIC NATURE OF DEDUCTION

Doubt has often been expressed as to whether a syllogism can add to our knowledge in any way. John Stuart Mill and Henri Poincaré, in particular, held the opinion that the conclusion of a syllogism is an “analytic” judgment in the sense of Kant, and therefore could be obtained by the mere dissection of the premisses. Any one, then, who maintains that mathematics is founded solely on logical principles would appear to maintain that mathematics, in the last instance, reduces to a huge tautology.

Mill, in Chapter III of Book II of his *System of Logic*, said that “it must be granted that in every syllogism, considered as an argument to prove the conclusion, there is a *petitio principii*. When we say

All men are mortal,
Socrates is a man,
therefore
Socrates is mortal,

it is unanswerably urged by the adversaries of the syllogistic theory, that the proposition, Socrates is mortal, is presupposed in the more general assumption, All men are mortal; that we cannot be assured of the mortality of all men unless we are already certain of the mortality of every individual man; that if it be still doubtful whether Socrates, or any other individual we choose to name, be mortal or not, the same degree of uncertainty must hang over the assertion, All men are mortal; that the general principle, instead of being given as evidence of the particular case, cannot itself be taken for true without exception until every shadow of doubt which could affect any case comprised with it is dispelled by evidence *aliunde*; and then what remains for the syllogism to prove? That, in short, no reasoning from general to particular can, as such, prove anything, since from a general principle we cannot infer any particulars but those which the principle itself assumes as known. This doctrine appears to me irrefragable....”

But it is not difficult to see that in certain cases at least deduction gives us *new knowledge*.^[56] If we already know that two and two always make four, and that Asquith and Lloyd George are two and so are the German Emperor and the Crown Prince, we can deduce that Asquith and Lloyd George and the German Emperor and the Crown Prince are four. This is new knowledge, not contained in our premisses, because the general proposition, “two and two are four,” never told us there were such people as Asquith and Lloyd George and the German Emperor and the Crown Prince, and the particular premisses did not tell us that there were four of them, whereas the particular proposition deduced does tell us both these things. But the newness of the knowledge is much less certain if we take the stock instance of deduction that is always given in books on logic, namely “All men are mortal; Socrates is a man, therefore Socrates is mortal.” In this case what we really know beyond reasonable doubt is that certain men, A, B, C, were mortal, since, in fact, they have died. If Socrates is one of these men, it is foolish to go the roundabout way through “all men are mortal” to arrive at the conclusion that *probably* Socrates is mortal. If Socrates is not one of the men on whom our induction is based, we shall still do better to argue straight from our A, B, C, to Socrates, than to go round by the general proposition, “all men are mortal.” For the probability that Socrates is mortal is greater, on our data, than the probability that all men are mortal. This is obvious, because if all men are mortal, so is Socrates; but if Socrates is mortal, it does not follow that all men are mortal. Hence we shall reach the conclusion that Socrates is mortal, with a greater approach to certainty if we make our argument purely inductive than if we go by way of “all men are mortal” and then use deduction.

Many years ago there appeared, principally owing to the initiative of Dr. F. C. S. Schiller of Oxford, a comic number of *Mind*. The idea was extraordinarily good, not so the execution. A German friend of Dr. Schiller was puzzled by the appearance of the advertisements, which were doubtfully humorous. However, by a syllogistic process, he acquired information which was new and useful to him, and thus incidentally refuted Mill. Presumably he started from the title of the magazine (*Mind!*), for a mark of exclamation seems nearly always in German to be a sign of an intended joke (including of course the mark after the politeness expressed in

the first sentence of a private letter or a public address). There would be, then, the following syllogism:

This is a book of would-be jokes (i.e. everything in this book is a would-be joke);

This advertisement is in this book;

Therefore, this advertisement is a would-be joke.

Thus the syllogism may be almost as powerful an agent in the detection of humour as M. Bergson's criterion, to be described in a future chapter. [\[57\]](#)

[\[56\]](#) [The following passage is almost word for word the same as a passage on pp. 123-5 of Mr. Russell's *Problems of Philosophy*, first published in 1912, a year after Mr. R*ss*ll's death. It is easy hastily to conclude that Mr. Russell was indebted to Mr. R*ss*ll to a greater degree than is usually supposed. But an examination of the internal evidence leads us to another conclusion. The two texts, it will be found, differ only in the names of the German Emperor, the Crown Prince and the other personages being replaced, in the book of 1912, by those of Messrs. Brown, Jones, Smith, and Robinson. Now, Mr. Russell, in a new edition of his *Problems* issued near the beginning of the European war and before the Russian revolution, substituted "the Emperor of Russia" for "the Emperor of China" of the first edition. Hence it seems quite likely that Mr. Russell, who has always shown a tendency to substitute existents for nonentities, wrote Mr. R*ss*ll's notes.—ED.]

[\[57\]](#) [See Chapter XLII.—ED.]

CHAPTER XXII

THE MORTALITY OF SOCRATES

The mortality of Socrates is so often asserted in books on logic that it may be as well briefly to consider what it means. The phrase “Socrates is mortal” may be thus defined: “There is at least one instant t such that t has not to Socrates the one-many relation R which is the converse of the relation ‘exists at,’ and all instants following t have not the relation R to Socrates, and there is at least one instant t' such that neither t' nor any instant preceding t' has the relation R to Socrates.”

This definition has many merits. In the first place, no assumption is made that Socrates ever lived at all. In the second place, no assumption is made that the instants of time form a continuous series. In the third place, no assumption is made as to whether Socrates had a first or last moment of his existence. If time be indeed a continuous series, then we can easily deduce^[58] that there must have been *either* a first moment of his non-existence *or* a last one of his existence, but not both; just as there seems to be either a greatest weight that a man can lift or a least weight that he cannot lift, but not both.^[59] This may be set forth as follows: for the present we will not concern ourselves with evidence for or against human immortality; I will merely try to present some logical questions which persistently arise whenever we think of eternal life. One of the greatest merits of modern logic is that it has allowed us to give precision to such problems, while definitely abandoning any pretensions of solving them; and I will now apply the logico-analytical method to one of the problems of our knowledge of the eternal world.^[60]

We will start from the generally accepted proposition that all men are mortal. Clearly, if we could know each individual man, and know that he was mortal, that would not enable us to know that all men are mortal, unless we knew, in addition, that those were all the men there are. But we need not here assume any such knowledge of general propositions; and, though most of us will admit that the proposition in question has great intrinsic plausibility, it is not strictly necessary for our present purpose to assume

anything more than the still more probable proposition “Socrates is mortal.” This last proposition, quite apart from the fact that we have a large amount of historical evidence for its truth, has been repeated so often in books on logic that it has taken on the respectable air of a platitude while preserving the character of an exceedingly probable truth. The truth also results from the fact that it is used as the conclusion of a syllogism. For it is a well-known fact that syllogisms can only be regarded as forming part of a sound education if the conclusions are obviously true. The use of a syllogism of the form “All cats are ducks and all ducks are mice, therefore all cats are mice,” would introduce grave doubts into the University of Oxford as to whether logic could any longer be considered as a valuable mental training for what are amusingly called the “learned professions.”

If, then, we divide all the instants of time, whether past, present, or future, into two series—those instants at which Socrates was alive, and those instants at which he was not alive—and leave out of consideration, for the sake of greater simplicity, all those instants before he lived, we see at once, by the simple application of Dedekind’s Axiom, that, if Socrates entered into eternal life after his death, there must have been either a last moment of his earthly life *or* a first moment of his eternal life, but not both.

Logic alone can give us no information as to which of these cases actually occurred, and we are thrown back on to a discussion of empirical evidence. It is no unusual thing to read of people who thought “that every moment would be their last.” In this case it is quite obvious that they consequently thought that eternity would have no beginning.

Now here we must consider two things: (1) It is plainly unsafe to conclude from what people think will happen to what will happen; (2) even if we could so conclude, it would be unsafe to deduce that there was a last moment in the life of Socrates: we could only make the guess plausible, as we should be using the inductive method.

There are two other pieces of evidence that there is a last moment of any earthly existence, which we may now briefly consider. That this was so was held by Carlo Michaelstaedter; but since he apparently only believed this because he wanted, by attributing a supposed ethical value to that moment, to give support to his theory of suicide, we ought not to give great weight to

this evidence. Secondly, Thomas Hobbes objected to the principle “that a quantity may grow less and less eternally, so as at last to be equal to another quantity; or, which is all one, that there is a last in eternity” as “void of sense.” Now, the principle meant is true, so that, although the other proposition mentioned by Hobbes does not follow logically from the first, there is some evidence that this other is true. In fact, that Hobbes thought that such-and-such a proposition followed from another proposition which he wrongly believed to be false, is far better evidence for the truth of such-and-such a proposition than any we have for the truth of most of our most cherished beliefs.

Thirdly, Leibniz, in a dialogue^[61] written on his journey of 1676 to visit Spinoza, raised the question whether the moment at which a man dies may be regarded as both the last moment at which he is alive and the first at which he is dead, as it must be by Aristotle’s theory of continuity. Agreement with this view violates the law of contradiction; denial of it implies that two moments can be immediately adjacent. By the denial, then, we are led to regard space and time as made up of indivisible points and moments, and thus, since we can draw one and only one parallel from any point in the diagonal of a square to a given side, the diagonal will contain the same (infinite) number of points as that side, and will therefore be equal to it. In this Leibniz repeated an argument used by the ancient Arabs, Roger Bacon, and William of Occam. This Leibniz considered to be a proof that a line cannot be an aggregate of points. Indeed, their number would be “the number of all numbers” of the greatest possible integer, which is not.

It does not seem, further, that any light is thrown on the logical question of human mortality or immortality by legal decisions. It would appear that one can, legally speaking, be alive for any period less than twenty-four hours after one is dead and be dead for any period less than twenty-four hours before one’s death. At least, according to *Salkeld*, i. 44, it was “adjudged that if one be born the first of February at eleven at night, and the last of January in the twenty-first year of his age, at one of the clock in the morning, he makes his will of lands, and dies, it is a good will, for he was then of age.” In Sir Robert Howard’s case (*ibid.*, ii. 625) it was held by Chief Justice Holt that “if A be born on the third day of September; and on the second day of September twenty-one years afterwards he make his will, this is a good will; for the law will make no fraction of a day, and by

consequence he was of age.” But it is hardly necessary to remark that in this way the problem with which we are concerned is merely shifted and not solved. For the question as to whether there is or is not a last moment of a man’s life is not answered by the decision that he dies legally twenty-four hours before or after he dies in the usual sense of the word, and the problem arises as to whether there is or is not a last moment of his legal age.^[62]

So assuming that there was a last moment of Socrates’s earthly life, and consequently no first moment of his eternal life, we see, further, that, unless the possibility of infinite numbers is granted, it would be quite possible for us logically to doubt the possibility of an eternal life for Socrates on the same grounds as those which led Zeno to assert that motion was impossible and that Achilles could never overtake the Tortoise. If, on the other hand, it be admitted that eternity, at least in the case of Socrates, had a beginning, these same arguments of Zeno would lead any one who denies the possibility of infinite number to conclude that Socrates, like the worm, can never die. Thus is it quite plain that the difficulties about immortality which meet us at the very outset of our inquiry can partly be solved only by the help of the theory of infinite numbers and partly, it would seem, not at all.

There is another difficulty about immortality which is quite distinct from this and is analogous to another argument of Zeno. If, indeed, all the instants of time be divided, as before, into the two series of instants at which Socrates was alive and instants at which he was not alive, it follows at once that no instant of time is not accounted for. At none of these instants, however, does Socrates die; obviously he cannot die either when he is alive or when he is dead. Thus it would appear that Socrates never died, and that we ought to re-define the term “mortal” to mean “a human being who is alive at some moments and dead at some.” Consequently we must avoid the very tempting conclusion that, because Socrates never died, he was therefore immortal.

It is very important carefully to distinguish between the two arguments I have just set forth. The second argument proves quite rigidly that Socrates and, indeed, anybody else, never dies, whether there is or is not a last moment of his life on earth. The first argument proves that, if there is a first moment of Socrates’s eternal life, his life on earth never ends. But we have

seen that we cannot conclude that this unending life proves that he never is or will be in a state of eternity.

[58] By “Dedekind’s Axiom,” *E. N.*, p. 11.

[59] *M.*, vol. xx., 1910, pp. 134-5.

[60] [Here, again, Mr. Russell’s work seems to anticipate some of Mr. Russell’s later work, e.g. in *Our Knowledge of the External World as a Field for Scientific Method in Philosophy*, Chicago and London, 1914, pp. 3-4, 55-6, *et passim*.—ED.]

[61] “Pacidius Philalethi” in Louis Couturat, *Opuscles et Fragments inédits de Leibniz*, Paris, 1903, pp. 594-627, especially pp. 599, 601, 608, 611. Cf. [A. E. Taylor, Hastings’ *Encyclopædia of Religion and Ethics*, vol. iv., Part 2, Edinburgh, 1912, p. 96.—ED.]; Robert Latta, *Leibniz: The Monadology and other Philosophical Writings*, Oxford, 1898, pp. 21 ff, 29 (note); Couturat, *La Logique de Leibniz d’après des documents inédits*, Paris, 1901, pp. 130, 132; and Russell, *Ph. L.*, pp. 108-16, 243-9.

[62] [It may be remarked that, according to *The Times* of December 20, 1917, Mr. Justice Sargant, in the Chancery Division, also held that “the law did not recognize fractions of a day,” and that Lord Blackburn, in his decision (9 *App. Cas.*, 371, 373) that a man born on the thirteenth of May 1853 attained the age of twenty-one on the thirteenth of May 1874 “was not speaking strictly.”—ED.]

CHAPTER XXIII

DENOTING

A concept *denotes* when, if it occurs in a proposition, the proposition is not about the concept, but *about* a term connected in a certain peculiar way with the concept. Some people often assert that man is mortal, and yet we never see announced in *The Times* that Man died on a certain day at his villa residence “Camelot” at Upper Tooting,^[63] nor do we hear that Procrastination was again the butt of Mr. Plowden’s jokes at Marylebone Police Court last week.

That two phrases may have different *meanings* and the same *denotation* was discovered by Alice and Frege. Alice^[64] observed that the road which led to Tweedledum’s house was that which led to the house of Tweedledee; and Frege pointed out that the phrases “the house to which the road that leads to Tweedledum’s house leads” and “the house to which the road that leads to Tweedledee’s house leads” have different *Sinn*, but the same *Bedeutung*.

^[63] Cf. *P. M.*, pp. 53-4.

^[64] See [Appendix M](#).

CHAPTER XXIV

THE

The word “the” implies existence and uniqueness; it is a mistake to talk of “the son of So-and-So” if So-and-So has a fine family of ten sons.^[65] People who refer to “the Oxford Movement” imply that Oxford only moved once; and those quaint people who say that “A is quite the gentleman” imply both the doubtful proposition that there is only one gentleman in the world, and the indubitably false proposition that he is that man. Probably A is one of those persons who add to the confusion in the use of the definite article by speaking of his wife as “the wife.”

In a certain Children’s Hymn Book one reads:

The river vast and small.

Few would deny that there is not more than one such river, but unfortunately it is doubtful if there is such a river at all. The case is exactly the same with the ontological proof of the existence of the most perfect being.^[66]

According to the *Daily Mail* of October 9, 1906, Judge Russell decided against a claim brought by an agent against his company for appointing another agent, the claim being on the ground that he was appointed as “the” agent.

Most people admit that the number 2 can be added to the number 2 to give the number 4, but this is a mistake. They concede, when they use *the*, that there is only one number 2, and yet they imagine that, when they consider it apart as the first term of our above sum, they can find another to add to it, and thereby form the third term. The truth is that “ $2 + 2 = 4$ ” is a very misleading equation, and what we really mean by that faultily abbreviated statement is more precisely: If x and y denote any things which form a class B, and x' and y' any other things that form a class (A) which, like that of x and y , is a member of the class (which we call “2”) of those classes which have a one-one correspondence with B (so that any member

of A corresponds to one, and only one, member of B, and conversely), the class of all the terms of A and B together is a member of that class of classes which, analogously, we call “4.” In this, for the sake of shortness, we have introduced abbreviations which should not be used in a rigorous logical statement.

[65] Cf. *Md.*, N. S., vol. xiv., 1905, pp. 481, 484.

[66] Cf. *ibid.*, p. 491, note.

CHAPTER XXV

NON-ENTITY

When people say that such-and-such a thing “is non-existent” they usually mean that there is not any “thing” of the kind spoken of. Venn meant this when he described^[67] his encounter with what he imagined to be a very ingenious tradesman: “I once had some strawberry plants furnished me which the vendor admitted would not bear many berries. But he assured me that this did not matter, since they made up in their size what they lost in their number. (He gave me, in fact, the hyperbolic formula, $xy = c$, to connect the number and magnitude.) When summer came, *no* fruit whatever appeared. I saw that it would be no use to complain, because the man would urge that the size of the non-existent berry was infinite, which I could not see my way to disprove. I had forgotten to bar zero values of either variable.”

It is to be regretted that this useful note was omitted in the second edition of Venn’s book; one can imagine that it might have protected Mr. MacColl and Herr Meinong (who believed, unlike Alice in what may be called her first theory,^[68] in round squares and fabulous monsters) against the dishonest practices of traders who were too ready with promises. For the death-blow to this kind of trade was not given until 1905, when Mr. Russell published his article “On Denoting,”^[69] and took up the position of the White King in opposition to Alice’s later assertions.^[70]

Venn’s experience illustrates another characteristic of mathematical logic. It is necessary, in order to make our arguments conclusive, to devote great care to the elimination of difficulties which rarely occur. The White Knight—who was like Boole in being a pioneer of mathematical logic in this way, and yet seems to have held, like Boole, those philosophical opinions which would base logic on psychology—recognized the necessity of taking precautions against any unusual appearance of mice on a horse’s back.^[71]

[67] *S. L.*, 1881, p. 339, note.

[68] See [Appendix N](#).

[69] *Md.*, N. S., vol. xiv., October 1905, pp. 479-93.

[70] See [Appendix N](#).

[71] See [Appendix O](#).

CHAPTER XXVI

IS

Is has four perfectly distinct meanings in English, besides misuses of the word. Among the misuses, perhaps the most important are those referred to by De Morgan:^[72] “... We say ‘murder *is* death to the perpetrator’ where the copula is *brings*; ‘two and two *are* four,’ the copula being ‘have the value of,’ etc.”

Schröder^[73] quite satisfactorily pointed out the well-known distinction between an *is* where subject and predicate can be interchanged (such as: “the class whose members are Shem, Ham and Japhet *is* the class of the sons of Noah”) and an *is* or *are* where they cannot (such as: Englishmen *are* Britons), but failed to see^[74] the more important distinction (made by Peano) of *is* in the sense of “is a member of.” If Englishmen *are* Britons, and Britons *are* civilized people, it follows that Englishmen *are* civilized people; but, though the *Harmsworth Encyclopædia* *is* a member of the class Book (of one or more volumes), and this class *is* the member of a class A of which it is the only member, yet the *Harmsworth Encyclopædia* *is* not a member of A, for it is not true that it is the whole class of books; and such a statement would not even be made except possibly in the form of an advertisement.

The fourth meaning of *is* is *exists*; it is in certain rare moods a matter for regret that there are difficulties in the way of using one word to denote four different things. For, if there were not, we might prove the existence of any thing we please by making it the subject of a proposition, and thereby earn the gratitude of theologians.

^[72] *F. L.*, p. 268.

^[73] *A. d. L.*, i. pp. 127 sqq.

^[74] *Ibid.*, vol. ii. pp. 461, 597.

CHAPTER XXVII

AND AND OR

When, with Boole, alternatives (A, B) are considered as mutually exclusive, logical addition may be described as the process of taking A *and* B or A *or* B. It is a great and rare convenience to have two terms for denoting the same thing: commonly, people denote several things by the same term, and only the Germans have the privilege of referring to, say, *continuity* as *Stetigkeit* or *Kontinuierlichkeit*. But Jevons^[75] quoted Milton, Shakespeare, and Darwin to prove that alternatives are not exclusive, and so attained first to recognized views by arguments which were plainly irrelevant.

Of course, *and* is often used as the sign of logical addition: thus one may speak of one's brothers *and* sisters, without being understood to mean the null-class (as should be the case), or pray for one's "relations and friends," without being sure that one's prayer would be answered,—as it certainly would if one meant to pray for the null-class, this being the class indicated. And a word like *while* is often used for a logical addition, when exclusiveness of the alternatives is almost implied. Thus, a reviewer in *Mind*,^[76] noticing the translation of Mach's *Popular Scientific Lectures* into American, said of the lectures that: "Most of them will be familiar ... to epistemologists and experimental psychologists: while the remainder, which deal with physical questions, are well worth reading." The reader has the impression, probably given unintentionally, that Professor Mach's epistemological and psychological lectures are not, in the reviewer's opinion, worth reading.

[75] *Pure Logic ...*, London, 1864, pp. 76-9. Cf. Venn, *S. L.*, 2nd ed., pp. 40-8.

[76] *N. S.*, vol. iv. p. 261.

CHAPTER XXVIII

THE CONVERSION OF RELATIONS

The “Conversion of Relations” does not mean what it might be supposed to mean; it has nothing to do with what Kant called “the wholesome art of persuasion.” What concerns us here is the convertibility of a logical relation. If A has a certain relation R to B, the relation of B to A, which may be denoted by $\check{R}$, is called the *converse* of R. As De Morgan^[77] remarked, this conversion may sometimes present difficulties. The following is De Morgan’s example:

“Teacher: ‘Now, boys, Shem, Ham and Japheth were Noah’s sons; who was the father of Shem, Ham and Japheth?’ No answer.

“Teacher: ‘Boys, you know Mr. Smith, the carpenter, opposite; has he any sons?’

“Boys: ‘Oh! yes, sir! there’s Bill and Ben.’

“Teacher: ‘And who is the father of Bill and Ben Smith?’

“Boys: ‘Why, Mr. Smith, to be sure.’

“Teacher: ‘Well, then, once more, Shem, Ham and Japheth were *Noah’s* sons; who was the father of Shem, Ham and Japheth?’

“A long pause; at last a boy, indignant at what he thought the attempted trick, cried out: ‘It *couldn’t* have been Mr. Smith.’ These boys had never converted the relation of father and son....”

[77] *Trans. Camb. Phil. Soc.*, vol. x., 1864, part ii., note on page 334.

CHAPTER XXIX

PREVIOUS PHILOSOPHICAL THEORIES OF MATHEMATICS

Mathematicians usually try to found mathematics on two principles:^[78] one is the principle of confusion between the sign and the thing signified (they call this principle the foundation-stone of the formal theory), and the other is the Principle of the Identity of Discernibles (which they call the principle of the permanence of equivalent forms).

But the truth is that if we set sail on a voyage of discovery with Logic alone at the helm, we must either throw such principles as “the identity of those conceptions which have in common the properties that interest us” and “the principle of permanence” overboard, or, if we do not like to act in such a way to old companions with whom we are so familiar that we can hardly feel contempt for them, at least recognize them clearly as having no logical validity and merely as psychological principles, and reduce them to the humble rank of stewards, to minister to our human weaknesses on the voyage. And then, if we adopt the wise policy of keeping our axioms down to the minimum number, we must refrain from creating or thinking that we are creating new numbers to fill up gaps among the older ones, and thence recognize that our rational numbers are not particular cases of “real” numbers, and so on.

We thus get a world of conceptions which looks, and is, very different from that which ordinary mathematicians think they see; and perhaps this is the reason why some mathematicians of great eminence, such as Hilbert and Poincaré, have produced such absurd discussions on the fundamental principles of mathematics,^[79] showing once more the truth of the not quite original remark of Aunt Jane, who

... observed, the second time
She tumbled off a 'bus:
“The step is short from the sublime
To the ridiculous.”

In their readiness to consider many different things as one thing—to consider, for example, the ratio 2:1 as the same thing as the cardinal number 2—such mathematicians as Peacock, Hankel, and Schubert were forestalled by the Pigeon, who thought that Alice and the Serpent were the same creature, because both had long necks and ate eggs.^[80] It is, however, doubtful whether the Pigeon would have followed the example of the mathematicians just mentioned so far as to embrace the creed of nominalism and so to feel no difficulty in subtracting from zero—a difficulty which was pointed out with great acuteness by the Hatter^[81] and modern mathematical logicians.

^[78] These principles, after many attempts to state them by Peacock, the Red and the White Queen (see [Appendix P](#)), Hankel, Schröder, and Schubert had been made, were first precisely formulated by Frege in *Z. S.*; cf. also Chapter VII.

^[79] See Couturat, *R. M. M.*, vol. xiv., March, 1906, pp. 208-50, and Russell, *ibid.*, September, 1906, pp. 627-34.

^[80] See [Appendix P](#).

^[81] See *ibid.*

CHAPTER XXX

FINITE AND INFINITE

I was once shown a statement made by an eminent mathematician of Cambridge from which one would conclude that this mathematician thought that finite distances became infinite when they were great enough. In one of those splendidly printed books, bound in blue, published by the University Press, and sold at about a guinea as a guide to some advanced branch of pure mathematics, one may read, even in the second edition published in 1900, the words: “Representation [of a complex variable] on a plane is obviously more effective for points at a finite distance from the origin than for points at a very great distance.”

Plainly some of the points at a very great distance are at a *finite* distance, for the same author mentions that Neumann’s sphere for representing the positions of points on a plane “has the advantage ... of exhibiting the uniqueness of $z = \infty$ as a value of the variable.”

CHAPTER XXXI

THE MATHEMATICAL ATTAINMENTS OF TRISTRAM SHANDY

Tristram Shandy^[82] said that his father was sometimes a gainer by misfortune; for if the pleasure of haranguing about it was as ten, and the misfortune itself only as five, he gained “half in half,” and was well off again as if the misfortune had never happened.

Suppose that the unit (arbitrary) of pleasure is denoted by A, Tristram Shandy, by neglecting, in this ethical discussion, to introduce negative quantities (Kant’s pamphlet advocating this introduction into philosophy was made subsequently^[83]), apparently made 15A to result, and this can hardly be maintained to be the half of 10A. It is possible, however, that Tristram Shandy succeeded in proving the apparently paradoxical equation

$$15A = 5A$$

by remarking that the axiom “the whole is greater than the part” does not always hold. This remark follows at once from what Mr. Russell^[84] has called “The Paradox of Tristram Shandy.” This paradox is described by Mr. Russell as follows:

“Tristram Shandy, as we know, took two years writing the history of the first two days of his life, and lamented that, at this rate, material would accumulate faster than he could deal with it, so that he could never come to an end. Now I maintain that, if he had lived for ever, and not wearied of his task, then, even if his life had continued as eventfully as it began, no part of his biography would have remained unwritten.”

This paradox is strictly correlative to the well-known paradox of Zeno about Achilles and the Tortoise.^[85] “The Achilles proves that two variables in a continuous series, which approach equality from the same side, cannot ever have a common limit: the Tristram Shandy proves that two variables which start from a common term, and proceed in the same direction, but

diverge more and more, may yet determine the same limiting class (which, however, is not necessarily a segment, because segments were defined as having terms beyond them). The Achilles assumes that whole and part cannot be similar, and deduces a paradox; the other, starting from a platitude, deduces that whole and part may be similar. For common-sense, it must be confessed that it is a most unfortunate state of things.” And Mr. Russell considers that, in the face of proofs, it ought to commit suicide in despair.

Now, I suggest the extremely unlikely possibility that Tristram Shandy, by reflection on his own life and literary labours, was led to the correct course of accepting the paradox which resulted from this reflection and rejecting the Achilles. Thus, he concluded that an infinite whole may be similar (or, in Cantor’s terminology, “equivalent”) to a proper part of itself, and hence, by a confusion of similarity with identity (or equivalence with equality) which he shares with some subsequent philosophers,^[86] that a whole may be equal to a proper part of itself. If A is an infinite class, it is not difficult to see that we can have

$$10A = 5A.$$

In this way many have avoided an opinion which rests on no better foundation than that formerly entertained by the inductive philosophers of Central Africa, that all men are black.^[87]

^[82] Cf. a letter of De Morgan in Mrs. De Morgan’s *Memoir of Augustus De Morgan*, p. 324.

^[83] Kant’s tract was published in 1763, while *Tristram Shandy* was published in 1760.

^[84] *P. M.*, pp. 358-9 [Cf. *M.*, vol. xxii., January 1912, p. 187.—ED.]

^[85] Cf. *P. M.*, pp. 350, 358-9; *M.*, vol. xxii., 1912, p. 157.

^[86] [Cf. for example, Cosmo Guastella, *Dell’ infinito*, Palermo, 1912.—ED.]

^[87] Cf. Russell, *P. M.*, p. 360.

CHAPTER XXXII

THE HARDSHIPS OF A MAN WITH AN UNLIMITED INCOME

I once heard a man refer to his income as limited, in order to illustrate the hardship of a class of men, of which he of course was one, in having to pay a somewhat high income-tax. It is obvious that this man spoke enviously, and consequently admitted the existence of more fortunately placed individuals who had unlimited incomes. A little reflection would have shown the man that he was not taking up a paradoxical attitude. A “paradoxical attitude” is of course the assertion of one or more propositions of which the truth cannot be perceived by a philosopher—and particularly an idealist—and can be perceived by a logician and occasionally, but not always, by a man of common-sense. Such propositions are: “The cat is hungry,” “Columbus discovered America,” and “A thing which is always at rest may move from the position A to the different position B.”

Now, if a man had an unlimited income, it is an immediate inference that, however low income-tax might be, he would have to pay annually to the Exchequer of his nation a sum equal in value to his whole income. Further, if his income was derived from a capital invested at a finite rate of interest (as is usual), the annual payments of income-tax would each be equal in value to the man’s whole capital. If, then, the man with an unlimited income chose to be discontented, he would be sure of a sympathetic audience among philosophers and business acquaintances; but discontent could not last long, for the thought of the difficulties he was putting in the way of the Chancellor of the Exchequer, who would find the drawing up of his budget most puzzling, would be amusing. Again, the discovery that, after paying an infinite income-tax, the income would be quite undiminished, would obviously afford satisfaction, though perhaps the satisfaction might be mixed with a slight uneasiness as to any action the Commissioners of Income-Tax might take in view of this fact.

A problem of a wholly different nature is connected with the possible purchase by the man with an unlimited income of an enumerable infinity of

pairs of boots. If he wished to prove that he had an even number of boots, it would be easy if right boots were distinguishable from left ones, but if the man were a faddist of such a kind that he insisted that his left boots should not be made in any way differently from his right ones, it would not be possible for him to prove the theorem mentioned unless he assumed what is known as “the multiplicative axiom.” In fact this axiom shows that it is legitimate to pick out an infinite succession of members of an infinite class in an arbitrary way. In the case of the pairs of boots, each pair contains two members, and if there is no means of distinguishing between them, when we wish to pick out one of them for each of the infinity of pairs, we cannot say which ones we mean to pick out unless we assume, by means of the above axiom, that a particularized member can always be found even with things of each of which it can be said that, like Private James in the *Bab Ballads*,

No characteristic trait had he
Of any distinctive kind.

However, a solution of the puzzle was given by Dr. Dénes König of Budapest. You first prove that there are points in space such that, if P is one of them, not more than a finite number of pairs of boots are such that each centre of mass of the two members of a pair is equidistant from P. Taking a point P of this sort, select from each pair the boot whose centre of mass is nearest P. (There may be a finite number of pairs left over, but they can be dealt with arbitrarily.)

Another form of the problem is as follows. Every time the man bought a pair of boots he also bought a pair of socks to go with it; he had an enumerable infinity of pairs of each, and the problem is to prove that he had as many boots as he had socks. In this case the boots, we will suppose, can be divided into right and left, but the socks cannot. Thus there are an enumerable infinity of boots, but the number of the socks cannot be determined without admitting the axiom mentioned above. A further difficulty might arise if the owner of the boots and socks lost one leg in some accident, and told his butler to give away half his socks. Naturally the butler would find great logical difficulties in so doing, and it would seem to be an interesting ethical problem whether he should be dismissed from his situation for failing to prove the multiplicative axiom. Again, if the butler

stole a pair of boots, the millionaire would have as many pairs as before, but might have fewer boots. There is as yet no evidence that the number of his boots is equal to or greater than the number of pairs.

CHAPTER XXXIII

THE RELATIONS OF MAGNITUDE OF CARDINAL NUMBERS

The theorems of cardinal arithmetic are frequently used in ordinary conversation. What is known as the Schröder-Bernstein theorem was used, long before Bernstein or Schröder, by Edward Thurlow, afterward the law-lord Lord Thurlow, when an undergraduate of Caius College, Cambridge. Thurlow was rebuked for idleness by the Master, who said to him: “Whenever I look out of the window, Mr. Thurlow, I see you crossing the Court.” The provost thus asserted a one-one correspondence between the class A of his acts of looking out of the window and a part of the class B of Thurlow’s acts of crossing the Court. Thurlow asserted in reply a one-one correspondence between B and a part of A: “Whenever I cross the Court I see you looking out of the window.” The Schröder-Bernstein theorem, then, allows us to conclude that there is a one-one correspondence between the classes A and B. That A and B were finite classes is not the fault of the Master or Thurlow; nor is it relevant logically.

CHAPTER XXXIV

THE UNKNOWNABLE

According to Mr. S. N. Gupta,^[88] the first thing that every student of Hindu logic has to learn when he is said to begin the study of inference is that “all H is S” is not always equivalent to “No H is not S.” “The latter proposition is an absurdity when S is *Kebalánvayi*, i.e. covers the whole sphere of thought and existence.... ‘Knowable’ and ‘Nameable’ are among the examples of *Kebalánvayi* terms. If you say there is a thing not-knowable, how do you know it? If you say there is a thing not-nameable, you must point that out, i.e. somehow name it. Thus you contradict yourself.”

Mr. Herbert Spencer’s doctrine of the “Unknowable” gives rise to some amusing thoughts. To state that all knowledge of such and such a thing is above a certain person’s intelligence is not self-contradictory, but merely rude: to state that all knowledge of a certain thing is above all possible human intelligence is nonsense, in spite of its modest, platitudinous appearance. For the statement seems to show that we do know something of it, viz. that it is unknowable.

To the last (1900) edition of *First Principles* was added a “Postscript to Part I,” in which the justice of this simple and well-known criticism as to the contradiction involved in speaking of an “Unknowable,” which had been often made during the forty odd years in which the various editions had been on the market, was grudgingly acknowledged as follows:^[89]

“It is doubtless true that saying what a thing is not, is, in some measure, saying what it is;... Hence it cannot be denied that to affirm of the Ultimate Reality that it is unknowable is, in a remote way, to assert some knowledge of it, and therefore involves a contradiction.”

The “Postscript” reminds one of the postscript to a certain Irishman’s letter. This Irishman, missing his razors after his return from a visit to a friend, wrote to his friend, giving precise directions where to look for the

missing razors; but, before posting the letter, added a postscript to the effect that he had found the razors.

One is tempted to inquire, analogously, what might be, in view of the Postscript, the point of much of Spencer's Part I. It is, to use De Morgan's^[90] description of the arguments of some who maintain that we can know nothing about infinity, of the same force as that of the man who answered the question how long he had been deaf and dumb.

But the best part of the joke against Mr. Spencer is that he, as we shall see in Chapter XXXVIII, was refuted by a fallacious argument, and thus mistakenly asserted the validity of the refutation of remarks which happen to be unsound.

The analogy of the contradiction of Burali-Forti with the contradiction involved in the notion of an "unknowable" may be set forth as follows. If A should say to B: "I know things which you never by any possibility can know," he may be speaking the truth. In the same way, ω may be said, without contradiction, to transcend all the *finite* integers. But if some one else, C, should say: "There are some things which no human being can ever know anything about," he is talking nonsense.^[91] And in the same way if we succeeded in imagining a number which transcends *all* numbers, we have succeeded in imagining the absurdity of a number which transcends itself.

All the paradoxes of logic (or "the theory of aggregates") are analogous to the difficulty arising from a man's statement: "I am lying."^[92] In fact, if this is true, it is false, and *vice versa*. If such a statement is spread out a little, it becomes an amusing hoax or an epigram. Thus, one may present to a friend a card bearing on both sides the words: "The statement on the other side of this card is false"; while the first of the epigrams derived from this principle seems to have been written by a Greek satirist:^[93]

Lerians are bad; not *some* bad and some *not*;
But all; there's not a Lorian in the lot,
Save Procles, that you could a good man call;—
And Procles—is a Lorian after all.

This is the original of a well-known epigram by Porson, who remarked that all Germans are ignorant of Greek metres,

All, save only Hermann;—
And Hermann's a German.

[88] *Md.*, N. S., vol. iv., 1895, p. 168.

[89] *First Principles*, 6th ed., 1900, pp. 107-10. The first edition was published in 1862.

[90] Note on p. 6 of his paper: "On Infinity; and on the Sign of Equality," *Trans. Camb. Phil. Soc.*, vol. xi., part i., pp. 1-45 (read May 16, 1864).

[91] The assertion of the finitude of a man's mind appears to be nonsense; both because, if we say that the mind of man is limited we tacitly postulate an "unknowable," and because, even if the human mind were finite, there is no more reason against its conceiving the infinite than there is for a mind to be blue in order to conceive a pair of blue eyes (cf. De Morgan, *loc. cit.*).

[92] Russell, *R. M. M.*, vol. xiv., September 1906, pp. 632-3, 640-4.

[93] *The Greek Anthology*, by Lord Neaves (Ancient Classics for English Readers), Edinburgh and London, 1897, p. 194.

CHAPTER XXXV

MR. SPENCER, THE ATHANASIAN CREED AND THE ARTICLES

When, in what I believe is misleadingly known as “The Athanasian Creed,” people say “The Father incomprehensible,” and so on, they are not falling into the same error as Mr. Spencer, for the Latin equivalent for “incomprehensible” is merely “*immensus*,” and Bishop Hilsey translated it more correctly as “immeasurable.”^[94] It is a regrettable fact that Dr. Blunt,^[95] in his mistaken modesty, has added a note to this passage that: “Yet it is true that a meaning not intended in the Creed has developed itself through this change of language, for the Nature of God is as far beyond the grasp of the mind as it is beyond the possibility of being contained within local bounds.”

Mr. Spencer seems no happier when we compare his statements with those in the Anglican Articles of Religion. There God is never referred to as infinite. It is true that His power and goodness are so referred to; but this deficiency was presumably brought about intentionally, so that faith might gain in meaning as time went on.

^[94] A. C. P., p. 217.

^[95] *Ibid.*, p. 218.

THE HUMOUR OF MATHEMATICIANS

Brahmagupta's problem^[96] appears to be the earliest instance of a kind of joke which has been much used by mathematicians. For the sake of giving a certain picturesqueness to the data of problems, and so to excite that sort of interest which is partly expressed by a smile, mathematicians have got into the habit of talking, for example, of monkeys in the form of geometrical points climbing up massless ropes. Professor P. Stäckel^[97] truly remarked that physiological mechanics—the mechanics of bones, muscles, and so on—is wholly different from this. There was once a lecturer on mathematics at Cambridge who used yearly to propound to his pupils a problem in rigid dynamics which related to the motion of a garden roller supposed to be without mass or friction, when a heavy and perfectly rough insect walked round the interior of it in the direction of normal rolling.

Hitherto this has been the only mathematical outlet for the humour of mathematicians; and those who really had the interests of mathematics at heart saw with alarm the growing tendency towards scholasticism in mathematical jokes. Fortunately the discovery of logic by some mathematicians has removed this danger. Still to many mathematicians logic is still unknown, and to them—to Professor A. Schoenflies for example—modern mathematics, owing to its alliance with logic, appears to be sinking into scholasticism. It is true that the word “scholasticism” is not used by Professor Schoenflies in any intentionally precise signification, but merely as a vague epithet of disapproval, as the word “socialism” is used by the ordinary philistine, and this would certainly serve as a sufficient excuse. But no excuse is needed: these opinions are themselves a source of mathematical jokes.

^[96] See Chapter XII.

^[97] *Encykl. der math. Wiss.*, vol. iv., part i., p. 474.

CHAPTER XXXVII

THE PARADOXES OF LOGIC

We have already^[98] referred to the contempt shown by some mathematicians for exact thought, which they condemn under the name of “scholasticism.” An example of this is given by Schoenflies in the second part of his publication usually known as the *Bericht über Mengenlehre*.^[99] Here^[100] a battle-cry in italics—

“Against all resignation, but also against all scholasticism!”—

found utterance. Later on, Schoenflies^[101] became bolder and adopted a more personal battle-cry, also in italics, and with a whole line to itself:

“For Cantorism but against Russellism!”

“Cantorism” means the theory of transfinite aggregates and numbers erected for the most part by Georg Cantor. Shortly speaking, the great sin of “Russellism” is to have gone too far in the chain of logical deduction for many mathematicians, who were perhaps, like Schoenflies,^[102] blinded by their rather uncritical love of mathematics. Thus it comes about that Schoenflies^[103] denounces Russellism as “scholastic and unhealthy.” This queer blend of qualities would surely arouse the curiosity of the most *blasé* as to what strange thing Russellism must be.^[104]

Schoenflies^[105] said that some mathematicians attributed to the logical paradoxes which have given Russell so much trouble to clear up, “especially to those that are artificially constructed, a significance that they do not have.” Yet no grounds were given for this assertion, from which it might be concluded that the rigid examination of any concept was unimportant. The paradoxes are simply the necessary results of certain logical views which are currently held, which views do not, except when they are examined rather closely, appear to contain any difficulty. The contradiction is not felt, as it happens, by people who confine their attention

to the first few number-classes of Cantor, and this seems to have given rise to the opinion, which it is a little surprising to find that some still hold, that cases not usually met with, though falling under the same concept as those usually met with, are of little importance. One might just as well maintain that continuous but not differentiable functions are unimportant because they are artificially constructed—a term which I suppose means that they do not present themselves when unasked for. Rather should we say that it is by the discovery and investigation of such cases that the concept in question can alone be judged, and the validity of certain theorems—if they are valid—conclusively proved. That this has been done, chiefly by the work of Russell, is simply a fact; that this work has been and is misunderstood by many^[106] is regrettable for this reason, among others, that it proves that, at the present time, as in the days in which *Gulliver's Travels* were written, some mathematicians are bad reasoners.^[107]

Nearly all mathematicians agreed that the way to solve these paradoxes was simply not to mention them; but there was some divergence of opinion as to how they were to be unmentioned. It was clearly unsatisfactory merely not to mention them. Thus Poincaré was apparently of opinion that the best way of avoiding such awkward subjects was to mention that they were not to be mentioned. But^[108] “one might as well, in talking to a man with a long nose, say: ‘When I speak of noses, I except such as are inordinately long,’ which would not be a very successful effort to avoid a painful topic.”

Schoenflies, in his paper of 1911 mentioned above, adopted the convenient plan of referring these logical difficulties at the root of mathematics to a department of knowledge which he called “philosophy.” He said^[109] of the theory of aggregates that though “born of the acuteness of the mathematical spirit, it has gradually fallen into philosophical ways, and has lost to some extent the compelling force which dwells in the mathematical process of conclusion.”

The majority of mathematicians have followed Schoenflies rather than Poincaré, and have thus adopted tactics rather like those of the March Hare and the Gryphon,^[110] who promptly changed the subject when Alice raised awkward questions. Indeed, the process of the first of these creatures of a child's dream is rather preferable to that of Schoenflies. The March Hare refused to discuss the subject because he was bored when difficulties arose.

Schoenflies would not say that he was bored—he professed interest in philosophical matters, but simply called the logical continuation of a subject by another name when he did not wish to discuss the continuation, and thus implied that he had discussed the whole subject. Further, Schoenflies would not apparently admit that the one method of logic could be applied to the solution of both mathematical and philosophical problems, in so far as these problems are soluble at all; but the March Hare, shortly before the remark we have just quoted, rightly showed great astonishment that butter did not help to cure both hunger and watches that would not go.^[111] The judgment of Schoenflies by which certain apparently mathematical questions were condemned as “philosophical,” rested on grounds as flimsy as those in the Dreyfus Case, or the Trial in *Wonderland*.^[112]

[98] Chapters VII and XXXVI.

[99] *Die Entwicklung der Lehre von den Punktmannigfaltigkeiten*. Bericht, erstattet der deutschen Mathematiker-Vereinigung, Leipzig, 1908.

[100] *Ibid.*, p. 7. The battle-cry is: “*Gegen jede Resignation, aber auch gegen jede Scholastik!*”

[101] “Ueber die Stellung der Definition in der Axiomatik,” *Jahresber, der deutsch. Math.-Ver.*, vol. xx., 1911, pp. 222-5. The battle-cry is on p. 256 and is: “Für den Cantorismus aber gegen den Russellismus!”

[102] *Ibid.*, p. 251. “Es ist also,” he exclaims with the eloquence of emotion and the emotion of eloquence, “nicht die Geringschätzung der Philosophie, die mich dabei treibt, sondern die Liebe zur Mathematik;...”

[103] “Ueber die Stellung der Definition in der Axiomatik,” *Jahresber, der deutsch. Math.-Ver.*, vol. xx., 1911, p. 251.

[104] [Cf. for this, *M.*, vol. xxii., January 1912, pp. 149-58.—ED.]

[105] *Bericht*, 1908, p. 76, note; cf. p. 72.

[106] E.g. in F. Hausdorff’s review of Russell’s *Principles* of 1903 in the *Vierteljahrsschr. für wiss. Philos. und Soziologie*.

[107] [Cf. *M.*, vol. xxv., 1915, pp. 333-8.—ED.]

[108] Russell, *A. J. M.*, vol. xxx., 1908, p. 226.

[109] *Loc. cit.*, p. 222.

[110] See [Appendix Q](#).

[111] See [Appendix R](#).

[112] See [Appendix S](#).

CHAPTER XXXVIII

MODERN LOGIC AND SOME PHILOSOPHICAL ARGUMENTS

The most noteworthy reformation of recent years in logic is the discovery and development by Mr. Bertrand Russell of the fact that the paradoxes—of Burali-Forti, Russell, König, Richard, and others—which have appeared of late years in the mathematical theory of aggregates and have just been referred to, are of an entirely *logical* nature, and that their avoidance requires us to take account of a principle which has been hitherto unrecognized, and which renders invalid several well-known arguments in refutation of scepticism, agnosticism, and the statement of a man that he asserts nothing.

Dr. Whitehead and Mr. Russell say:^[113] “The principle which enables us to avoid illegitimate totalities may be stated as follows: ‘Whatever involves *all* of a collection must not be one of the collection,’ or conversely: ‘If, provided a certain collection had a total, it would have members only definable in terms of that total, then the said collection has no total.’ We shall call this the ‘vicious-circle principle,’ because it enables us to avoid the vicious circles involved in the assumption of illegitimate totalities. Arguments which are condemned by the vicious-circle principle will be called ‘vicious-circle fallacies.’ Such arguments, in certain circumstances, may lead to contradictions, but it often happens that the conclusions to which they lead are in fact true, though the arguments are fallacious. Take, for example, the law of excluded middle in the form ‘all propositions are true or false.’ If from this law we argue that, because the law of excluded middle is a proposition, therefore the law of excluded middle is true or false, we incur a vicious-circle fallacy. ‘All propositions’ must be in some way limited before it becomes a legitimate totality, and any limitation which makes it legitimate must make any statement about the totality fall outside the totality. Similarly the imaginary sceptic who asserts that he knows nothing and is refuted by being asked if he knows that he knows nothing, has asserted nonsense, and has been fallaciously refuted by an

argument which involves a vicious-circle fallacy. In order that the sceptic's assertion may become significant it is necessary to place some limitation upon the things of which he is asserting his ignorance; the proposition that he is ignorant of every member of this collection must not itself be one of the collection. Hence any significant scepticism is not open to the above form of refutation."

In fact, the world of things falls into various sets of things of the same "type." For every propositional function $\phi(x)$ there is a range of values of x for which $\phi(x)$ has a signification as a true or a false proposition. Until this theory was brought forward, there were occasionally discussions as to whether an object which did not belong to the range of a certain propositional function possessed the corresponding property or not. Thus, Jevons, in early days,^[114] was of opinion that virtue is neither black nor not-black because it is not coloured, but rather later^[115] he admitted that virtue is not triangular.^[116]

^[113] *Pa. Ma.*, p. 40.

^[114] *S. o. S.* pp. 36-7.

^[115] *E. L. L.*, pp. 120-1.

^[116] [It may perhaps be added that, some years after Mr. Russell's death, Dr. Whitehead stated, in an address delivered in 1916 and reprinted in his book on *The Organisation of Thought* (London, 1917, p. 120), that "the specific heat of virtue is 0.003 is, I should imagine, not a proposition at all, so that it is neither true nor false...."—ED.]

CHAPTER XXXIX

THE HIERARCHY OF JOKES

Jokes may be divided into various types. Thus a joke or class of jokes can only be the subject of a joke of higher order. Otherwise we would get the same vicious-circle fallacy which gives rise to so many paradoxes in logic and mathematics. A certain Oxford scholar succeeded, to his own satisfaction, in reducing all jokes to primitive types, consisting of thirty-seven proto-Aryan jokes. When any proposition was propounded to him, he would reflect and afterwards pronounce on the question as to whether the proposition was a joke or not. If he decided, by his theory, that it was a joke, he would solemnly say: “There *is* that joke.” If this narration is accepted as a joke, since it cannot be reduced to one of the proto-Aryan jokes under pain of leading us to commit a vicious-circle fallacy, we must conclude that there is at least one joke which is not proto-Aryan; and, in fact, is of a higher type. There is no great difficulty in forming a hierarchy of jokes of various types. Thus a joke of the fourth type (or order) is as follows: A joke of the first order was told to a Scotchman, who, as we would expect, was unable to see it.^[117] The person (A) who told this joke told the story of how the joke was received to another Scotchman thereby making a joke about a joke of the first order, and thus making a joke of the second order. A remarked on this joke that no joke could penetrate the head of the Scotchman to whom the joke of the first order was told, even if it were fired into his head with a gun. The Scotchman, after severe thought, replied: “But ye couldn’t do that, ye know!” A repeated the whole story, which constituted a joke of the third order, to a third Scotchman. This last Scotchman again, after prolonged thought, replied: “He had ye there!” This whole story is a joke of the fourth order.

Most known jokes are of the first order, for the simple reason that the majority of people find that the slightest mental effort effectually destroys any perception of humour. It seems to me that a joke becomes more pleasurable in proportion as logical faculties are brought into play by it; and hence that logical power is allied, or possibly identical, with the power of grasping more subtle jokes. The jokes which amuse the frequenters of

music-halls, Conservatives, and M. Bergson—and which usually deal with accidents, physical defects, mothers-in-law, foreigners, or over-ripe cheese—are usually jokes of the first order. Jokes of the second, and even of the third, order appeal to ordinary well-educated people; jokes of higher order require either special ability or a sound logical training on the part of the hearer if the joke is to be appreciated; while jokes of transfinite order presumably only excite the inaudible laughter of the gods.

[\[117\]](#) [It may be that, like certain remarks about cheese and mothers-in-law (see below), the statement that Scotchmen cannot see jokes is a joke of the first order.—ED.]

CHAPTER XL

THE EVIDENCE OF GEOMETRICAL PROPOSITIONS

It has often been maintained that the twentieth proposition of the first book of Euclid—that two sides of a triangle are together greater than the third side—is evident even to asses. This does not, however, seem to me generally true. I once asked a coastguardsman the distance from A to B; he replied: “Eight miles.” On further inquiry I elicited the fact that the distance from A to C was two miles and the distance from C to B was twenty-two miles. Now the paths from A to B and from C to B were by sea; while the path from A to C was by land. Hence if the path by land was rugged and the distance along the road was two miles, it would appear that the coastguardsman believed that not only could one side of a triangle be greater than the other two, but that one straight side of a triangle might be greater than one straight side and any curvilinear side of the same triangle. The only escape from part of this astonishing creed would be by assuming that the distance of two miles from A to C was measured “as the crow flies,” while the road A to C was so hilly that a pedestrian would traverse more than fourteen miles when proceeding from A to C. Then indeed the coastguardsman could maintain the true proposition that there is at least one triangle ABC, with the side AC curvilinear, such that the sum of the lengths of AB and AC is greater than the length of BC, and only deny the twentieth proposition of the first book of Euclid.

Reasoning with the coastguardsman only had the effect of his adducing the authority of one Captain Jones in support of the accuracy of his data. Possibly Captain Jones held strange views as to the influence of temperature or other physical circumstances, or even the nature of space itself, on the lengths of lines in the neighbourhood of the triangle ABC.

ABSOLUTE AND RELATIVE POSITION

Some people maintain that position in space or time must be relative because, if we try to determine the position of a body A, if bodies B, C, D with respect to which the position of A could be determined were not present, we should be trying to determine something about A without having our senses affected by other things. These people seem to me to be like the cautious guest who refused to say anything about his host's port-wine until he had tasted red ink.

“Wherein, then,” says Mr. Russell,^[118] “lies the plausibility of the notion that all points are exactly alike? This notion is, I believe, a psychological illusion, due to the fact that we cannot remember a point so as to know it when we meet again. Among simultaneously presented points it is easy to distinguish; but though we are perpetually moving, and thus being brought among new points, we are quite unable to detect this fact by our senses, and we recognize places only by the objects they contain. But this seems to be a mere blindness on our part—there is no difficulty, so far as I can see, in supposing an immediate difference between points, as between colours, but a difference which our senses are not constructed to be aware of. Let us take an analogy: Suppose a man with a very bad memory for faces; he would be able to know, at any moment, whether he saw one face or many, but he would not be aware whether he had seen any of the faces before. Thus he might be led to define people by the rooms in which he saw them, and to suppose it self-contradictory that new people should come to his lectures, or that old people should cease to do so. In the latter point at least it will be admitted by lecturers that he would be mistaken. And as with faces, so with points—inability to recognize them must be attributed, not to the absence of individuality, but merely to our incapacity.”

Another form of this tendency is shown by Kronecker, Borel, Poincaré, and many other mathematicians, who refuse mere logical determination of a conception and require that it be actually described in a finite number of terms. These eminent mathematicians were anticipated by the empirical

philosopher who would not pronounce that the “law of thought” that A is either in the place B or not is true until he had looked to make sure. This philosopher was of the same school as J. S. Mill and Buckle, who seem to have maintained implicitly not only that, in view of the fact that the breadth of a geometrical line depends upon the material out of which it is constructed, or upon which it is drawn, that there ought to be a paste-board geometry, a stone geometry, and so on;^[119] but also that the foundations of logic are inductive in their nature.^[120] “We cannot,” says Mill,^[121] “conceive a round square, not merely because no such object has ever presented itself in our experience, for that would not be enough. Neither, for anything we know, are the two ideas in themselves incompatible. To conceive a body all black and yet white would only be to conceive two different sensations as produced in us simultaneously by the same object—a conception familiar to our experience—and we should probably be as well able to conceive a round square as a hard square, or a heavy square, if it were not that in our uniform experience, at the instant when a thing begins to be round, it ceases to be square, so that the beginning of the one impression is inseparably associated with the departure or cessation of the other. Thus our inability to form a conception always arises from our being compelled to form another contradictory to it.”

^[118] *Md.*, N. S., vol. x., July, 1901, pp. 313-14.

^[119] J. B. Stallo, *The Concepts and Theories of Modern Physics*, 4th ed., London, 1900, pp. 217-27.

^[120] *Ibid.*, pp. 140-4.

^[121] *Examination of the Philosophy of Sir William Hamilton*, vol. i. p. 88, Amer. ed.

CHAPTER XLII

LAUGHTER

[It seemed advisable to give here^[122] some views on laughter, most of which were also held by Mr. R*ss*ll, though no written expression of his views has yet been found. In a review^[123] of M. Bergson's book on *Laughter*,^[124] Mr. Russell has remarked:

“It has long been recognized by publishers that everybody desires to be a perfect lady or gentleman (as the case may be); to this fact we owe the constant stream of etiquette-books. But if there is one thing which people desire even more, it is to have a faultless sense of humour. Yet so far as I know, there is no book called ‘Jokes without Tears, by Mr. McQuedy.’ This extraordinary lacuna has now been filled. Those to whom laughter has hitherto been an unintelligible vagary, in which one must join, though one could never tell when it would break out, need only study M. Bergson's book to acquire the finest flower of Parisian wit. By observing a very simple formula they will know infallibly what is funny and what is not; if they sometimes surprise their unlearned friends, they have only to mention their authority in order to silence doubt. ‘The attitudes, gestures and movements of the human body,’ says M. Bergson, ‘are laughable in exact proportion as that body reminds us of a mere machine.’ When an elderly gentleman slips on a piece of orange-peel and falls, we laugh, because his body follows the laws of dynamics instead of a human purpose. When a man falls from a scaffolding and breaks his neck on the pavement, we presumably laugh even more, since the movement is even more completely mechanical. When the clown makes a bad joke for the first time, we keep our countenance, but at the fifth repetition we smile, and at the tenth we roar with laughter, because we begin to feel him a mere automaton. We laugh at Molière's misers, misanthropists and hypocrites, because they are mere types mechanically dominated by a master impulse. Presumably we laugh at Balzac's characters for the same reason; and presumably we never smile at Falstaff, because he is individual throughout.”

The review concludes with the reflection that “it would seem to be impossible to find any such formula as M. Bergson seeks. Every formula treats what is living as if it were mechanical, and is therefore by his own rule a fitting object of laughter.” Now, this undoubtedly true conclusion has been obtained, as is readily seen, by a vicious-circle fallacy which Mr. R*ss*ll would hardly have committed.—ED.]

[122] From a remark on p. 47 above, it is evident that Mr. R*ss*ll intended to write some such chapter as this.

[123] *The Professor’s Guide to Laughter*, *The Cambridge Review*, vol. xxxii., 1912, pp. 193-4.

[124] *Laughter, an Essay on the Meaning of the Comic*, English translation by C. Brereton and F. Rothwell, London, 1911.

CHAPTER XLIII

“GEDANKENEXPERIMENTE” AND EVOLUTIONARY ETHICS

The “Gedankenexperimente,” upon which so much weight has been laid by Mach^[125] and Heymans,^[126] had already been investigated by the White Queen,^[127] who, however, seems to have perceived that the results of such experiments are not always logically valid. The psychological founding of logic appears to be not without analogy with the surprising method of advocates of evolutionary ethics, who expect to discover what *is* good by inquiring what cannibals have *thought* good. I sometimes feel inclined to apply the historical method to the multiplication table. I should make a statistical inquiry among school-children, before their pristine wisdom had been biassed by teachers. I should put down their answers as to what 6 times 9 amounts to, I should work out the average of their answers to six places of decimals, and should then decide that, at the present stage of human development, this average is the value of 6 times 9.

[125] See, e.g., *E. u. I.*, pp. 183-200.

[126] *G. u. E.*, vol. i.

[127] See [Appendix T](#).

APPENDIXES

A. LOGIC AND THE PRINCIPLE OF IDENTITY.

T. L. G., p. 45: ““Contrariwise,” continued Tweedledee, “if it was so, it might be; and if it were so, it would be: but as it isn’t, it ain’t. That’s logic.”

S. B., p. 159: The Professor said: “The day is the same length as anything that is the same length as *it*.”

S. B., p. 161: Bruno observed that, when the Other Professor lost himself, he should shout: “He’d be sure to hear hisself, ‘cause he couldn’t be far off.”

B. SYNTHESIS OF CONTRADICTIONS.

T. L. G., p. 71: ““What a beautiful belt you’ve got on!’ Alice suddenly remarked.... ‘At least,’ she corrected herself on second thoughts, ‘a beautiful cravat, I should have said—no, a belt, I mean—I beg your pardon!’ she added in dismay, for Humpty-Dumpty looked thoroughly offended, and she began to wish she hadn’t chosen that subject. ‘If only I knew,’ she thought to herself, ‘which was neck and which was waist!’”

C. EMPIRICAL PHILOSOPHERS AND MATHEMATICS.

T. L. G., p. 79: ““... Now if you had the two eyes on the same side of the nose, for instance—or the mouth at the top—that would be *some* help.’

“‘It wouldn’t look nice,’ Alice objected. But Humpty-Dumpty only shut his eyes and said: ‘Wait till you’ve tried.’”

T. L. G., p. 72: “‘And if you take one from three hundred and sixty-five, what remains?’

“‘Three hundred and sixty-four, of course.’

“‘Humpty-Dumpty looked doubtful. ‘I’d rather see that done on paper,’ he said.”

D. NOMINAL DEFINITION.

T. L. G., p. 73: “‘When *I* used a word,’ Humpty-Dumpty said in rather a scornful tone, ‘it means just what I choose it to mean—neither more nor less.’

“‘The question is,’ said Alice, ‘whether you *can* make words mean different things.’

“‘The question is,’ said Humpty-Dumpty, ‘which is to be master—that’s all.’”

E. CONFORMITY OF A PARADOXICAL LOGIC WITH COMMON-SENSE.

T. L. G., p. 100:

“But I was thinking of a plan
To dye one’s whiskers green,
And always use so large a fan
That they could not be seen.”

(Verse from White

Knight’s song.)

F. IDEALISTS AND THE LAWS OF LOGIC.

T. L. G., p. 52-3: Tweedledee exclaimed: “‘... if he [the Red King] left off dreaming about you [Alice], where do you suppose you’d be?’

“‘Where I am now, of course,’ said Alice.

“‘Not you!’ Tweedledee retorted contemptuously. ‘You’d be nowhere. Why, you’re only a sort of thing in his dream!’

“‘If that there King was to wake,’ added Tweedledum, ‘you’d go out—bang!—just like a candle!’

“‘I shouldn’t!’ Alice exclaimed indignantly. ‘Besides, if *I’m* only a sort of thing in his dream, what are *you*, I should like to know?’

“‘Ditto,’ said Tweedledum...; ‘you know very well you’re not real.’

“‘I *am* real!’ said Alice, and began to cry.”

T. L. G., p. 97: “‘How *can* you go on talking so quietly, head downwards?’ Alice asked, as she dragged him out by the feet, and laid him in a heap on the bank.

“‘The Knight looked surprised at the question. ‘What does it matter where my body happens to be?’ he said. ‘My mind goes on working all the same. In fact, the more head downwards I am, the more I keep inventing new things.’”

T. L. G., p. 98: “‘... Everybody that hears me sing—either it brings the *tears* into their eyes, or else——’

“‘Or else what?’ said Alice, for the Knight had made a sudden pause.

“‘Or else it doesn’t, you know.’”

G. DISTINCTION BETWEEN SIGN AND SIGNIFICATION.

T. L. G., pp. 98-9: “‘The name of the song is called “*Haddocks’ Eyes*.”’

“‘Oh, that’s the name of the song, is it?’ Alice said, trying to feel interested.

“‘No, you don’t understand,’ the Knight said looking a little vexed. ‘That’s what the name is *called*. The name really is “*The Aged Aged Man*.”’

“‘Then I ought to have said “That’s what the *song* is called”?’ Alice corrected herself.

“‘No, you oughtn’t: that’s another thing. The *song* is called “*Ways and Means*”: but that’s only what it’s *called*, you know!’

“‘Well, what *is* the song, then?’ said Alice, who was by this time completely bewildered.

“‘I was coming to that,’ the Knight said. ‘The song really is “*A-sitting on a Gate*”....’”

H. NOMINALISM.

A. A. W., p. 70: “‘Then you should say what you mean,’ the March Hare went on.

“‘I do,’ Alice hastily replied; ‘at least—at least I mean what I say—that’s the same thing, you know.’

“‘Not the same thing a bit!’ said the Hatter. ‘Why, you might just as well say that “I see what I eat” is the same thing as “I eat what I see.”’

“‘You might just as well say,’ added the March Hare, ‘that “I like what I get” is the same thing as “I get what I like”!’

“‘You might just as well say,’ added the Dormouse, which seemed to be talking in its sleep, ‘that “I breathe when I sleep” is the same as “I sleep when I breathe”!’

“‘It *is* the same thing with you,’ said the Hatter; and here the conversation dropped,...

I. UTILITY OF SYMBOLIC LOGIC.

A. A. W., p. 92: “‘I quite agree with you,’ said the Duchess, ‘and the moral of that is—“Be what you would seem to be”—or if you’d like it put more simply—“Never imagine yourself not to be otherwise than what it might appear to others that what you were or might have been was not otherwise than what you had been would have appeared to them to be otherwise.”’

“‘I think I should understand that better,’ Alice said very politely, ‘if I had it written down: but I can’t quite follow it as you say it.’

“‘That’s nothing to what I could say if I chose,’ the Duchess replied, in a pleased tone.”

J. MISTAKE AS TO THE NATURE OF CRITICISM.

T. L. G., p. 105: “‘She’s in that state of mind,’ said the White Queen, ‘that she wants to deny *something*—only she doesn’t know what to deny.’

“‘A nasty, vicious temper,’ the White Queen remarked; and then there was an uncomfortable silence for a minute or two.”

K. A CRITERION OF TRUTH.

H. S., p. 3:

“Just the place for a Snark! I have said it twice:
That alone should encourage the crew.
Just the place for a Snark! I have said it thrice:
What I tell you three times is true.”

H. S., p. 50:

“‘Tis the note of the Jubjub! Keep count. I entreat;
You will find I have told it you twice.

'Tis the song of the Jubjub! The proof is complete,
If only I've stated it thrice."

L. UNIVERSAL AND PARTICULAR PROPOSITIONS.

T. L. G., p. 40: The Gnat had told Alice that the Bread-and-butterfly lives on weak tea with cream in it; so:

"“Supposing it couldn't find any?’ she suggested.

"“Then it would die, of course.’

"“But that must happen very often,’ Alice remarked thoughtfully.

"“It always happens,’ said the Gnat.”

M. DENOTING.

T. L. G., p. 43: Tweedledum and Tweedledee were, in many respects, indistinguishable, and Alice, walking along the road, noticed that “whenever the road divided there were sure to be two finger-posts pointing the same way, one marked ‘TO TWEEDLEDUM’S HOUSE’ and the other ‘TO THE HOUSE OF TWEEDLEDEE.’

"“I do believe,’ said Alice at last, ‘that they live in the same house!...’”

N. NON-ENTITY.

T. L. G., p. 87: “‘I always thought they [human children] were fabulous monsters!’ said the Unicorn....

"“Do you know [said Alice], I always thought Unicorns were fabulous monsters, too! I never saw one alive before!’

"“Well, now that we *have* seen each other,’ said the Unicorn, ‘if you’ll believe in me, I’ll believe in you. Is that a bargain?’”

T. L. G., pp. 80-1: “‘I see nobody on the road,’ said Alice.

“‘I only wish *I* had such eyes,’ the [White] King remarked in a fretful tone. ‘To be able to see Nobody! And at that distance, too! Why, it’s as much as *I* can do to see real people by this light!’”

A. A. W., p. 17: “And she [Alice] tried to fancy what the flame of a candle looks like after the candle is blown out, for she could not remember ever having seen such a thing.”

A. A. W., p. 68: “... This time it [the Cheshire Cat] vanished quite slowly, beginning with the end of the tail, and ending with the grin, which remained some time after the rest of it had gone.

“‘Well! I’ve often seen a cat without a grin,’ thought Alice; ‘but a grin without a cat! It’s the most curious thing I ever saw in all my life!’”

A. A. W., p. 77: “... The Dormouse went on,...; ‘and they drew all manner of things—everything that begins with an M.’

“‘Why with an M?’ said Alice.

“‘Why not?’ said the March Hare.

“Alice was silent.

“... [The Dormouse] went on: ‘—that begins with an M, such as mouse-traps, and the moon, and memory, and muchness, you know you say things are “much of a muchness”—did you ever see such a thing as a drawing of a muchness?’

“‘Really, now you ask me,’ said Alice, very much confused, ‘I don’t think _____,’

“‘Then you shouldn’t talk,’ said the Hatter.”

O. OBJECTS OF MATHEMATICAL LOGIC.

T. L. G., p. 93: “‘I was wondering what the mouse-trap [fastened to the White Knight’s saddle] was for,’ said Alice. ‘It isn’t very likely there would be any mice on the horse’s back.’

“‘Not very likely, perhaps,’ said the Knight, ‘but, if they *do* come, I don’t choose to have them running all about.’

“‘You see,’ he went on after a pause, ‘it’s as well to be provided for *everything*. That’s the reason the horse has all these anklets round his feet.’

“‘But what are they for?’ Alice asked in a tone of great curiosity.

“‘To guard against the bites of sharks,’ the Knight replied.”

P. THE PRINCIPLE OF PERMANENCE.

T. L. G., p. 106: “‘Can you do Subtraction? [said the Red Queen] Take nine from eight.’

“‘Nine from eight I can’t, you know,’ Alice replied very readily ‘but—’

“‘She can’t do Substraction,’ said the White Queen.”

A. A. W., p. 56: [Said the Pigeon to Alice]: “‘... No, no! You’re a serpent; and there’s no use denying it. I suppose you’ll be telling me next that you never tasted an egg!’

“‘I *have* tasted eggs certainly,’ said Alice, who was a very truthful child; ‘but little girls eat eggs quite as much as serpents do, you know.’

“‘I don’t believe it,’ said the Pigeon; ‘but if they do, why then they’re a kind of serpent, that’s all I can say.’

“This was such a new idea to Alice, that she was quite silent for a minute or two, which gave the Pigeon the opportunity of adding, ‘You’re looking

for eggs, I know *that* well enough; and what does it matter to me whether you're a little girl or a serpent?

“‘It matters a good deal to *me*,’ said Alice hastily;...”

A. A. W., p. 75: “‘But why [asked Alice] did they live at the bottom of a well?’

“‘Take some more tea,’ the March Hare said to Alice, very earnestly.

“‘I’ve had nothing yet,’ Alice replied in an offended tone, ‘so I can’t take more.’

“‘You mean you can’t take *less*,’ said the Hatter: ‘it’s very easy to take *more* than nothing.’”

Q. MATHEMATICIANS’ TREATMENT OF LOGIC.

A. A. W., p. 74: The Hatter had told of his quarrel with Time, and of Time’s refusal now to do anything he asked: “‘... It’s always six o’clock now!’

“‘A bright idea came into Alice’s head. ‘Is that the reason so many tea things are put out here?’ she asked.

“‘Yes, that’s it,’ said the Hatter, with a sigh: ‘it’s always tea time, and we’ve no time to wash the things between whiles.’

“‘Then you keep moving round, I suppose?’ said Alice.

“‘Exactly so,’ said the Hatter: ‘as the things get used up.’

“‘But what happens when you come to the beginning again?’ Alice ventured to ask.

“‘Suppose we change the subject,’ the March Hare interrupted, yawning. ‘I’m getting tired of this.’”

A. A. W., p. 99: “‘And how many hours a day did you do lessons?’ said Alice, in a hurry to change the subject.

“‘Ten hours the first day,’ said the Mock Turtle, ‘nine the next, and so on.’

“‘What a curious plan!’ exclaimed Alice.

“‘That’s the reason they’re called lessons,’ the Gryphon remarked, ‘because they lessen from day to day.’

“This was quite a new idea to Alice, and she thought it over a little before she made her next remark. ‘Then the eleventh day must have been a holiday.’

“‘Of course it was,’ said the Mock Turtle.

“‘And how did you manage on the twelfth?’ Alice went on eagerly.

“‘That’s enough about lessons,’ the Gryphon interrupted in a very decided tone....”

R. METHOD IN MATHEMATICS AND LOGIC.

A. A. W., p. 71: “‘Two days wrong!’ sighed the Hatter. ‘I told you butter wouldn’t suit the works!’ he added, looking angrily at the March Hare.

“‘It was the *best* butter,’ the March Hare meekly replied.

“‘Yes, but some crumbs must have got in as well,’ the Hatter grumbled; ‘you shouldn’t have put it in with the bread-knife.’

“The March Hare took the watch and looked at it gloomily: then he dipped it into his cup of tea, and looked at it again: but he could think of nothing better to say than his first remark, ‘It was the *best* butter, you know.’”

S. VERDICT THAT LOGIC IS PHILOSOPHY.

A. A. W., pp. 119-23: "... 'Consider your verdict,' he [the King] said to the jury, in a low trembling voice.

"'There's more evidence to come yet, please your Majesty,' said the White Rabbit, jumping up in a great hurry: 'this paper has just been picked up.'

"'What's in it?' said the Queen.

"'I haven't opened it yet,' said the White Rabbit, 'but it seems to be a letter written by the prisoner to—to somebody.'

"'It must have been that,' said the King, 'unless it was written to nobody, which isn't usual, you know.'

"'Who is it directed to?' said one of the jurymen.

"'It isn't directed at all,' said the White Rabbit, 'in fact there's nothing written on the *outside*.' He unfolded the paper as he spoke, and added, 'It isn't a letter, after all: it's a set of verses.'

"'Are they in the prisoner's handwriting?' asked another of the jurymen.

"'No they're not,' said the White Rabbit, 'and that's the queerest thing about it.' (The jury all looked puzzled).

"'He must have imitated somebody else's hand,' said the King. (The jury brightened up again.)

"'Please your Majesty,' said the Knave, 'I didn't write it, and they can't prove that I did: there's no name signed at the end.'

"'If you didn't sign it, said the King, that only makes the matter worse. You *must* have meant some mischief, or else you'd have signed your name like an honest man.'

"There was a general clapping of hands at this: it was the first really clever thing the King had said that day.

"'That *proves* his guilt, of course,' said the Queen, 'so, off with——'

“‘It doesn’t prove anything of the sort!’ said Alice. ‘Why, you don’t even know what they’re about!’

“‘Read them,’ said the King.

“The White Rabbit put on his spectacles. ‘Where shall I begin, please your Majesty?’ he asked.

“‘Begin at the beginning,’ the King said very gravely, ‘and go on till you come to the end: then stop.’

“There was dead silence in the court, whilst the White Rabbit read out these verses:

*“‘They told me you had been to her,
And mentioned me to him;
She gave me a good character,
But said I could not swim.*

*He sent them word I had not gone
(We know it to be true):
If she should push the matter on,
What would become of you?*

*I gave her one, they gave him two,
You gave us three or more;
They all returned from him to you,
Though they were mine before.*

*If I or she should chance to be
Involved in this affair,
He trusts to you to set them free
Exactly as they were.*

*My notion was that you had been
(Before she had this fit)
An obstacle that came between
Him, and ourselves, and it.*

*Don't let him know she liked them best,
For this must ever be
A secret kept from all the rest,
Between yourself and me.'*

“‘That’s the most important piece of evidence we’ve heard yet,’ said the King, rubbing his hands, ‘so now let the jury——’

“‘If any one of them can explain it,’ said Alice (she had grown so large in the last few minutes that she wasn’t a bit afraid of interrupting him), ‘I’ll give him sixpence. *I don’t believe there’s an atom of meaning in it.*’

“The jury all wrote down on their slates, ‘She doesn’t believe there’s an atom of meaning in it,’ but none of them attempted to explain the paper.

“‘If there’s no meaning in it,’ said the King, ‘that saves a world of trouble, you know, as we needn’t try to find any. And yet I don’t know,’ he went on, spreading out the verses on his knee and looking at them with one eye; ‘I seem to see some meaning in them after all. “— *said I could not swim*”; you can’t swim, can you?’ he added, turning to the Knave.

“The Knave shook his head sadly. ‘Do I look like it?’ he said. (Which he certainly did *not*, being made entirely of cardboard.)

“‘All right, so far,’ said the King; and he went on muttering over the verses to himself: “*We know it to be true*’—that’s the jury, of course—‘*If she should push the matter on*’—that must be the Queen—‘*What would become of you?*’ What indeed!—‘*I gave her one, they gave him two!*’ why, that must be what he did with the tarts, you know——’

“‘But it goes on, ‘*They all returned from him to you,*’” said Alice.

“‘Why, there they are!’ said the King, triumphantly pointing to the tarts on the table. ‘Nothing can be clearer than that. Then again—‘*Before she had this fit*’—you never had fits, my dear, I think?’ he said to the Queen.

“‘Never!’ said the Queen furiously, throwing an inkstand at the Lizard as she spoke. (The unfortunate little Bill had left off writing on his slate with one finger, as he found it made no mark; but he now hastily began again, using the ink that was trickling down his face, as long as it lasted.)

“‘Then the words don’t *fit* you,’ said the King, looking round the court with a smile. There was a dead silence.

“‘It’s a pun!’ the King added in an angry tone, and everybody laughed.

“‘Let the jury consider their verdict,’ the King said, for about the twentieth time that day.

“‘No, no!’ said the Queen. ‘Sentence first—verdict afterwards.’

“‘Stuff and nonsense!’ said Alice loudly. ‘The idea of having the sentence first!’

“‘Hold your tongue!’ said the Queen, turning purple....”

T. “GEDANKENEXPERIMENTE.”

T. L. G., p. 61: “Alice laughed. ‘There’s no use trying,’ she said: ‘one *can’t* believe impossible things.’

“‘I daresay you haven’t had much practice,’ said the [White] Queen. ‘When I was your age, I always did it for half-an-hour a day. Why, sometimes I’ve believed as many as six impossible things before breakfast.’”

Printed in Great Britain by
UNWIN BROTHERS, LIMITED, THE GRESHAM PRESS, WOKING AND LONDON

*** END OF THE PROJECT GUTENBERG EBOOK THE
PHILOSOPHY OF MR. B*RT*ND R*SS*LL ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.